Supplementary Information

Distal Chemical Shift Perturbations Reveal That Ligand Induced Reshaping of Protein Conformational Landscapes is Ubiquitous

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Supplementary Text

The CSPdb

The CSPdb includes receptors that regulate diverse biological processes, including cell cycle progression (e.g., MAD2: 1KLQ), transcriptional control (e.g., Nrd1: 2MOW, 6GC3), translational regulation, autophagy, apoptosis (e.g., BCL-XL family: 2M04, 2LP8), and cell adhesion (e.g., tensin: 2LOZ). Disease-relevant targets address mechanisms in leukemia and other cancers, Alzheimer's disease (e.g., amyloid precursor protein-binding proteins: 2ROZ), and infectious diseases from viruses and bacteria, such as anthrax (2RUI), chikungunya (5I22), SARS-CoV-2 (7PKU), HIV (2L6E), and Ebola (2KQ0).

The CSPdb features several recurring protein domains that exhibit notable conformational changes upon ligand binding. For instance, it includes 23 interactions involving calmodulin (CaM) or CaM-like proteins with peptides (e.g., PDB IDs: 1SY9, 2JZI; full list in **SI Table S1**), which often involve substantial rearrangements upon binding described in detail elsewhere[1]. Other receptors known for binding-induced conformational shifts or folding include the G protein Cdc42 (1CF4), the metaphase chromosome spindle checkpoint protein MAD2 (1KLQ), the KIX domain (2LXS), and CREB-binding proteins (e.g., 1R8U, 1L8C).

Additional recurring domains include the extra-terminal (ET) domains of bromodomain proteins BRD3 and BRD4 (e.g., 2NCZ, 2ND0; full list in **SI Table S7**, **SI Fig. S6**), which bind intrinsically-disordered regions of viral integrase or chromatin regulator proteins. These interactions trigger subtle changes distal from the binding site, including the formation of a stabilizing secondary structure that reorients two allosteric helices[2, 3]. Similarly, the PDZ2 domain (e.g., 1D5G, 1VJ6, 2M0V) shows distal effects mediated by the $\beta2$–$\beta3$ loop during peptide recognition[4].

Nine entries involve the PH domain of the TFIIH transcription initiation factor subunit Tfb1 (e.g., 2LOX, 2M14; full list in **SI Table S8**, **SI Fig. S7**). In these complexes, the ligand is often modeled as a helical polypeptide bound orthogonally to an extended β-sheet, resulting in signal propagation via hydrogen-bond rearrangements. Other targets which involve hydrogen bond rearrangement in β-sheets include the Cdc42 (1CF4), Cyclophilin A (2MS4), and SH3 domain (e.g., 2OJ2, 2ROL; full list in **SI Table S1**). An additional eight entries use ubiquitin or ubiquitin-like (e.g. SUMO and NEDD8) protein receptors. Ubiquitin has been used as a model system for NMR studies[5] and for studying allostery[6] (e.g., 2KWV, 2MUR; full list in **SI Table S9**, **SI Fig. S8**).

NMR ring-current contributions to chemical shift perturbations

Based on the r^{-3} distance dependence of aromatic ring-current effects and established ring-current parameters for aromatic amino acids, the theoretical maximum distance for a ≥ 0.05-ppm HN ring-current shift is approximately 10 Å for Phe/Tyr and 12 Å for Trp under idealized geometry[7, 8].

Estimated maximum distance for a ≥ 0.05-ppm amide ^{1}H ring-current shift.

Aromatic ring-current contributions to ^{1}H chemical shifts can be approximated using the equivalent-dipole model,

$$\Delta\delta_{\mathrm{RC}} = \mu \frac{(3\cos^2\theta - 1)}{r^3},$$

where r is the distance from the proton to the aromatic ring center, θ is the angle between this vector and the normal to the ring plane, and μ is the effective ring-current constant. Abraham et al. determined $\mu = 26.2\,\mathrm{ppm\,Å^3}$ for benzene[7]. The maximum absolute effect occurs at $\theta = 0°$ or 180° corresponding to the proton lying along the normal above or below the aromatic ring plane, where the angular term has a magnitude of 2. Using the 0.05-ppm significance threshold gives

$$r_{\max} = \left(\frac{2\mu}{0.05}\right)^{1/3} = 10.2\,\text{Å}.$$

Thus, even under idealized geometry, a Phe/Tyr-like aromatic ring would need to be within approximately 10 Å of an amide proton to produce a significant ring-current effect. For Trp, the contributions from its two aromatic rings are summed; using the relative ring-current intensities reported by Perkins et al. gives an idealized upper-bound estimate of approximately 12 Å.

NMR electric-field induced chemical shift perturbations

Based on the r^{-2} distance dependence of electric-field effects and experimentally determined parameters for protein amide chemical shifts, a one-unit charge change is estimated to produce a ≥ 0.05-ppm HN shift only within approximately 6–8 Å under idealized conditions[9, 10].

Estimated maximum distance for a ≥ 0.05-ppm amide ^{1}H electric-field induced shift.

Electric-field-induced amide chemical shifts can be approximated using the Buckingham treatment,

$$|\Delta\delta_{\mathrm{HN}}| = |A|\,|E|\,|\cos\beta|,$$

where A is the HN shielding polarizability, E is the change in electric field at the amide proton, and β is the angle between these vectors. Using $|A| = 21\,\mathrm{ppm\,\AA^2}/e$ for backbone amide protons[10] and approximating the electric field from a point-charge change using Coulomb's law gives

$$|\Delta\delta_{\mathrm{HN}}| = \frac{21\,|\Delta q|}{\epsilon_{\mathrm{eff}}\,r^2}\,|\cos\beta|.$$

For a one-unit charge change ($|\Delta q| = 1e$), optimal orientation ($|\cos\beta| = 1$), and the 0.05-ppm significance threshold,

$$r_{\mathrm{max}} = \left[\frac{21}{0.05\,\epsilon_{\mathrm{eff}}}\right]^{1/2}.$$

Experimentally derived effective dielectric constants for proteins are approximately 6–13; values of ∼6.5–7 and 8.6 ± 0.8 have been obtained from protein CSP measurements[9, 10]. These values correspond to idealized maximum distances of approximately 6–8 Å for a ≥ 0.05-ppm ^{1}HN shift from a one-unit charge change.

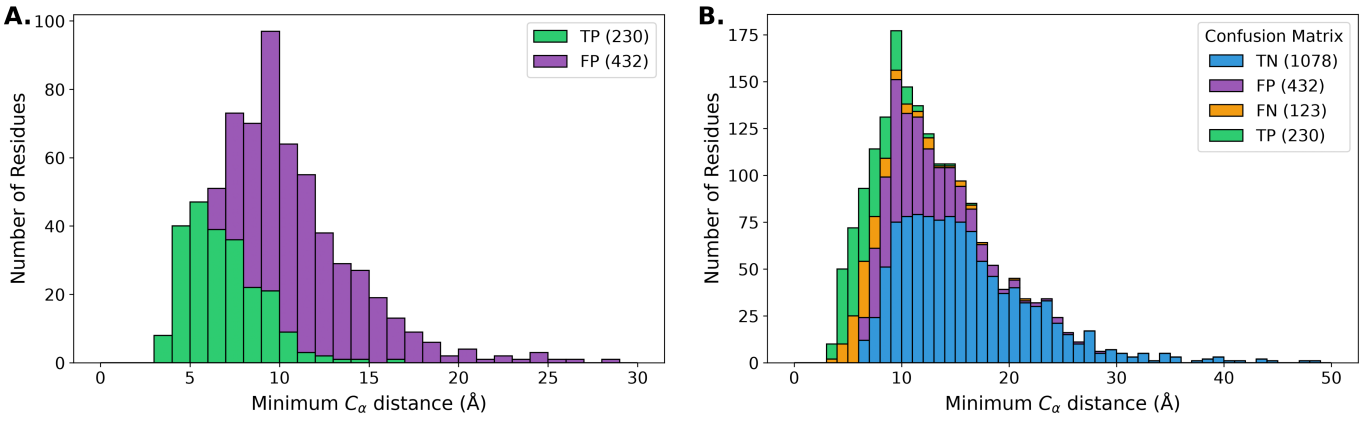
Unlike aromatic ring-current effects, a universal distance cutoff cannot be assigned to electrostatic CSPs because their magnitude depends on charge, orientation, dielectric environment, and screening. Small electric-field-induced ^{1}HN CSPs have been experimentally detected > 12 Å from a charge perturbation, although these were below the 0.05-ppm significance threshold used here[10]. Although our apo and holo datasets were matched for pH, ligand binding itself may alter protein ionization states and thereby generate electric-field-induced CSPs.

SI Table S1. Data for all receptors in this study

apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC	apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
	1YWI	6558	6559	0.78	0.72		2KC8	16066	16065	0.52	0.25
	2MCN	16641	19447	0.79	0.71		2KYL	15972	16967	0.42	0.25
	2N9P	25913	25914	0.72	0.57		6U19	16620	27875	0.46	0.24
	2L5E	50148	17270	0.58	0.54		7PKU	50446	34661	0.42	0.24
	6RH6	34394	34395	0.56	0.53		2LGK	17813	17812	0.52	0.24
	6GC3	17173	34260	0.52	0.52		2K7L	27288	15919	0.39	0.24
	2M0G	18802	18808	0.48	0.52		1H8B	17627	4453	0.33	0.24
	2PLD	5318	5310	0.59	0.52		6H8C	18827	34307	0.31	0.23
	2ROL	15710	11040	0.52	0.52		1KLQ	52275	5299	0.33	0.23
	2LI5	16835	17879	0.57	0.51		2MUR	25229	25230	0.57	0.23
	5I22	4871	30010	0.56	0.50		2L4T	17254	17255	0.31	0.22
	2ROZ	10235	10236	0.59	0.50		2N3A	34537	25639	0.37	0.22
7JMY	7JYZ	30782	30791	0.56	0.48		2JZI	51289	15624	0.43	0.22
	6IVU	36220	36221	0.67	0.48		2CZY	27227	6749	0.38	0.21
	2LBM	15001	17569	0.60	0.45		2IPA	6075	15028	0.19	0.21
	2K7A	5461	16809	0.50	0.45		7SFT	36097	30958	0.41	0.21
	2KSP	15279	16671	0.53	0.44		2KZU	17018	17019	0.25	0.20
2LSY	2N1G	18455	25559	0.53	0.44		2RR4	11362	11115	0.40	0.20
	6FDP	19757	34223	0.48	0.43		2YS5	11095	11094	0.30	0.20
	2MOW	17173	19954	0.41	0.43		2LOZ	10318	17364	0.31	0.19
2LSY	2LSK	18455	18434	0.48	0.43		2FIN	6809	7024	0.19	0.19
	2L6E	16555	17307	0.55	0.43	1U2N	1R8U	6268	5987	0.56	0.19
	2YKA	16697	17693	0.43	0.42	6VED	2L3R	30703	17200	0.31	0.18
2JNS	6BNH	15125	30373	0.53	0.42		2JW1	15503	15504	0.29	0.18
7JMY	7JQ8	30782	30786	0.54	0.41		5J8H	51289	30063	0.43	0.17
	1N7T	18785	5631	0.45	0.40		2LP8	18250	18238	0.20	0.17
	2LVO	18581	18582	0.56	0.40		2NMB	4683	4263	0.19	0.17
7JMY	7JYN	30782	30790	0.51	0.40		2LXS	6095	18694	0.32	0.17
	2JXC	4184	15554	0.60	0.39		1VJ6	5131	6060	0.30	0.16
	6FDT	19757	34224	0.42	0.38		1PD7	6899	5808	0.38	0.16
	2OJ2	4122	6581	0.42	0.37		8SG2	27579	31080	0.37	0.16
	2N8T	25866	25865	0.61	0.37		2LE8	16396	17702	0.22	0.15
1MM2	2L75	5555	17344	0.65	0.36		2L0Y	15763	17067	0.30	0.15
	6OQJ	30599	30605	0.48	0.36		1ONV	27288	5685	0.35	0.14
	6G04	34247	34248	0.44	0.35		2M0K	51289	17771	0.42	0.14
	1DDM	4683	4263	0.33	0.35		2HUG	6592	7241	0.38	0.14
2JNS	2NCZ	15125	26041	0.46	0.35		5U5S	50149	30206	0.24	0.14
	2KPL	6911	16559	0.43	0.35	1V49	2N9X	5958	25919	0.28	0.13
	7X5C	51441	36473	0.42	0.35	2KQ3	2KHS	16585	15357	0.34	0.13
	9QLM	15670	34986	0.60	0.34		2KTB	51472	16694	0.20	0.13
	2LP0	25142	17407	0.43	0.34		6QXZ	27250	27251	0.42	0.13
	2MC6	19428	19429	0.46	0.33		2KWI	15524	15525	0.40	0.13
	2M0G	19034	18808	0.44	0.33		2RUI	16811	11570	0.17	0.10
1W1F	1WA7	6261	6456	0.48	0.32		6U4N	17827	30659	0.15	0.10
	2RQG	4915	10019	0.44	0.32		5MF9	34068	34067	0.21	0.09
	9R3Y	26643	34992	0.47	0.31		2KOH	27205	16507	0.26	0.06
	2FFK	6809	7024	0.30	0.31		2MJV	15057	19738	0.29	0.06
	2B0F	5659	6823	0.31	0.31		1SY9	51289	5480	0.36	0.06
	2M0U	18824	18825	0.35	0.31		2N80	25883	25829	0.27	0.04
	2BN5	6691	6690	0.28	0.30		2G35	15792	7061	0.17	0.04
	2L29	17128	17127	0.27	0.30		2JMX	6564	15072	0.08	0.04
	2MUR	17769	25230	0.40	0.30		2MNZ	19913	19914	0.47	0.04
	6E5N	25544	30500	0.40	0.30	2JNS	2ND0	15125	26042	0.12	0.03
1SSF	2MWO	5878	25347	0.30	0.29		2M0J	51289	17771	0.35	0.02
	2KGI	16209	16210	0.62	0.29	2JNS	2ND1	15125	26043	0.11	0.02
2JNS	6BGG	30782	30367	0.32	0.29		2LGG	17813	17808	0.38	-0.03
	2N3K	15125	25649	0.43	0.29		2I94	5332	7293	0.15	-0.04
1SSF	2LVM	5878	18579	0.33	0.28		2MES	51289	19238	0.30	-0.05
	2M5B	50942	19045	0.32	0.28		2KVM	15674	16778	0.33	-0.07
	7KLR	30808	30809	0.55	0.28		5OEO	51289	34161	0.17	-0.07
	7S5J	30926	30950	0.40	0.27		2K8F	34231	15944	0.16	-0.08
1SSF	2MWP	5878	25348	0.27	0.27		2LSP	15057	18439	0.14	-0.08
	6R5G	28070	34384	0.40	0.27		2LGF	51289	17807	0.28	-0.10
	5VF0	17769	30276	0.31	0.27		2N55	52209	25694	0.24	-0.12
	2K6Q	5958	15877	0.37	0.26		9C5E	18642	31177	0.22	-0.17
	2K17	15670	15671	0.54	0.26		2JMF	6262	15016	0.00	-0.19
7QCX	1D5G	34688	4516	0.27	0.26		2K6D	52081	15866	0.08	-0.19
	2LXM	18681	18682	0.28	0.26						

SI Table S2. Data for Hydrolase Receptors

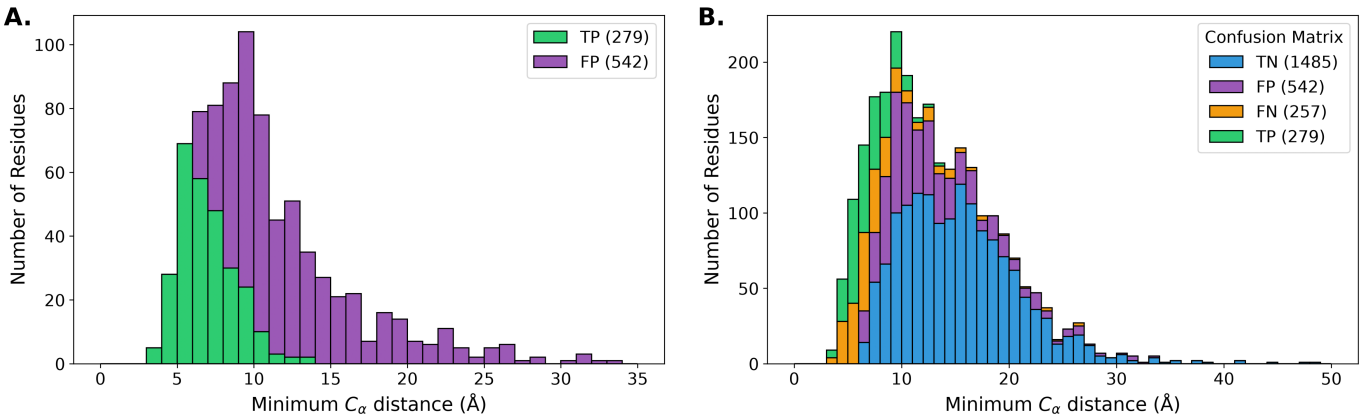
apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC	apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
	6GC3	17173	34260	0.52	0.52		2KYL	15972	16967	0.42	0.25
	2PLD	5318	5310	0.59	0.52		2K7L	27288	15919	0.39	0.24
	2LBM	15001	17569	0.60	0.45		2JZI	51289	15624	0.43	0.22
	6FDP	19757	34223	0.48	0.43		2LOZ	10318	17364	0.31	0.19
1MM2	2L75	5555	17344	0.65	0.36		1VJ6	5131	6060	0.30	0.16
	6OQJ	30599	30605	0.48	0.36		2LE8	16396	17702	0.22	0.15
	2RQG	4915	10019	0.44	0.32		1ONV	27288	5685	0.35	0.14
2JNS	6BGG	30782	30367	0.32	0.29		2HUG	6592	7241	0.38	0.14
	6R5G	28070	34384	0.40	0.27	2KQ3	2KHS	16585	15357	0.34	0.13
7QCX	1D5G	34688	4516	0.27	0.26		2KWI	15524	15525	0.40	0.13
	2KC8	16066	16065	0.52	0.25						



SI Fig. S1. Confusion Matrix Histograms for Hydrolase Receptors ($n = 21$)

SI Table S3. Data for Transferase Receptors

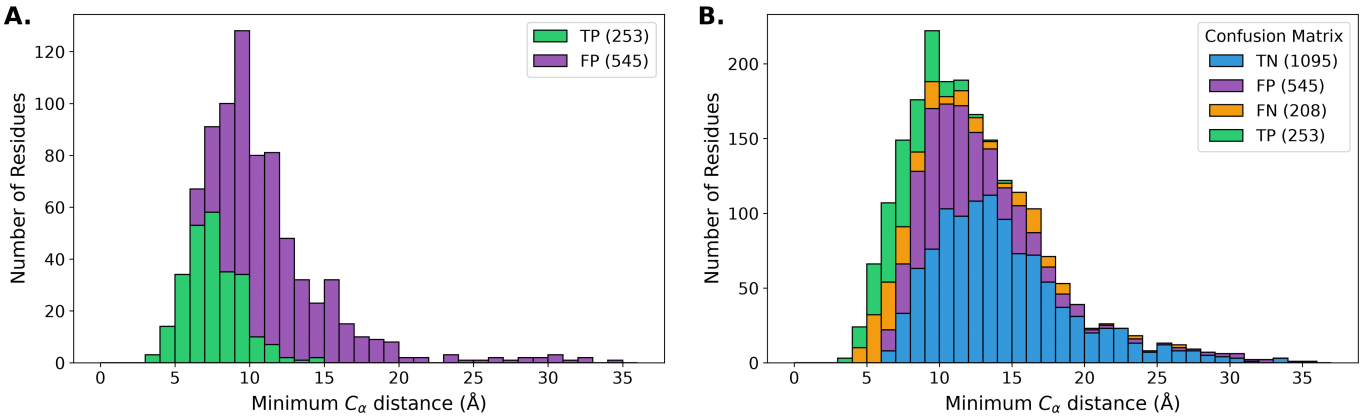
apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC	apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
	2N9P	25913	25914	0.72	0.57		2LGK	17813	17812	0.52	0.24
	2PLD	5318	5310	0.59	0.52		2YS5	11095	11094	0.30	0.20
	2K7A	5461	16809	0.50	0.45	1U2N	1R8U	6268	5987	0.56	0.19
2LSY	2N1G	18455	25559	0.53	0.44	6VED	2L3R	30703	17200	0.31	0.18
	2MOW	17173	19954	0.41	0.43		5J8H	51289	30063	0.43	0.17
2LSY	2LSK	18455	18434	0.48	0.43		2LXS	6095	18694	0.32	0.17
	2LVO	18581	18582	0.56	0.40		8SG2	27579	31080	0.37	0.16
7JMY	7JYN	30782	30790	0.51	0.40		6QXZ	27250	27251	0.42	0.13
	2OJ2	4122	6581	0.42	0.37		2G35	15792	7061	0.17	0.04
	2N8T	25866	25865	0.61	0.37	2JNS	2ND1	15125	26043	0.11	0.02
2JNS	2NCZ	15125	26041	0.46	0.35		2LGG	17813	17808	0.38	-0.03
	2MC6	19428	19429	0.46	0.33		2I94	5332	7293	0.15	-0.04
1W1F	1WA7	6261	6456	0.48	0.32		2K8F	34231	15944	0.16	-0.08
	5VF0	17769	30276	0.31	0.27		9C5E	18642	31177	0.22	-0.17
	2KYL	15972	16967	0.42	0.25		2JMF	6262	15016	0.00	-0.19
	6U19	16620	27875	0.46	0.24						



SI Fig. S2. Confusion Matrix Histograms for Transferase Receptors ($n = 31$)

SI Table S4. Data for All Alpha Receptors

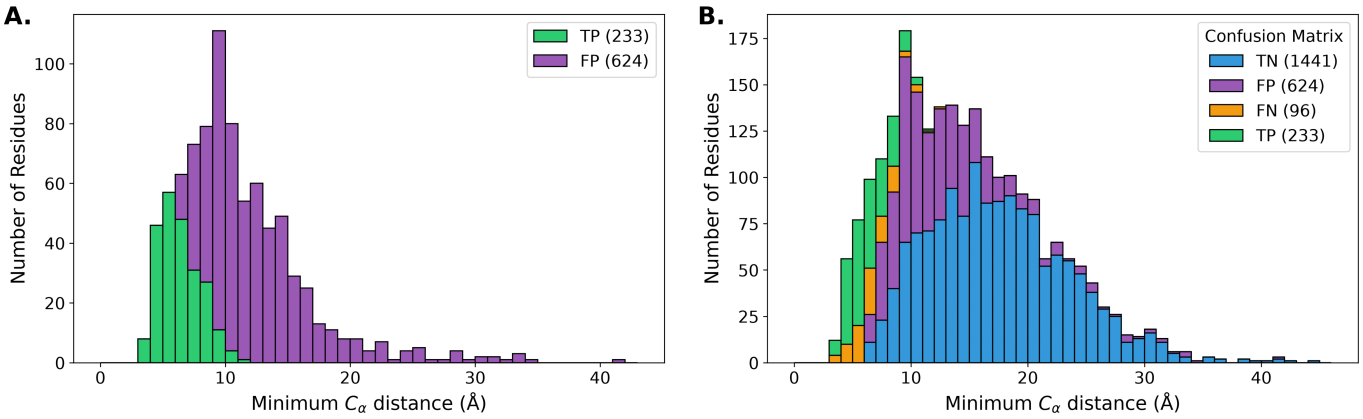
apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC	apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
	2KSP	15279	16671	0.53	0.44		2LXS	6095	18694	0.32	0.17
	6FDP	19757	34223	0.48	0.43		1PD7	6899	5808	0.38	0.16
	2L6E	16555	17307	0.55	0.43		1ONV	27288	5685	0.35	0.14
	2JXC	4184	15554	0.60	0.39		2M0K	51289	17771	0.42	0.14
	6FDT	19757	34224	0.42	0.38		2KTB	51472	16694	0.20	0.13
	2LP0	25142	17407	0.43	0.34		1SY9	51289	5480	0.36	0.06
	2RQG	4915	10019	0.44	0.32		2N80	25883	25829	0.27	0.04
	1H8B	17627	4453	0.33	0.24		2M0J	51289	17771	0.35	0.02
	2JZI	51289	15624	0.43	0.22		2I94	5332	7293	0.15	-0.04
	5J8H	51289	30063	0.43	0.17		2MES	51289	19238	0.30	-0.05



SI Fig. S3. Confusion Matrix Histograms for All Alpha Receptors ($n = 20$)

SI Table S5. Data for All Beta Receptors

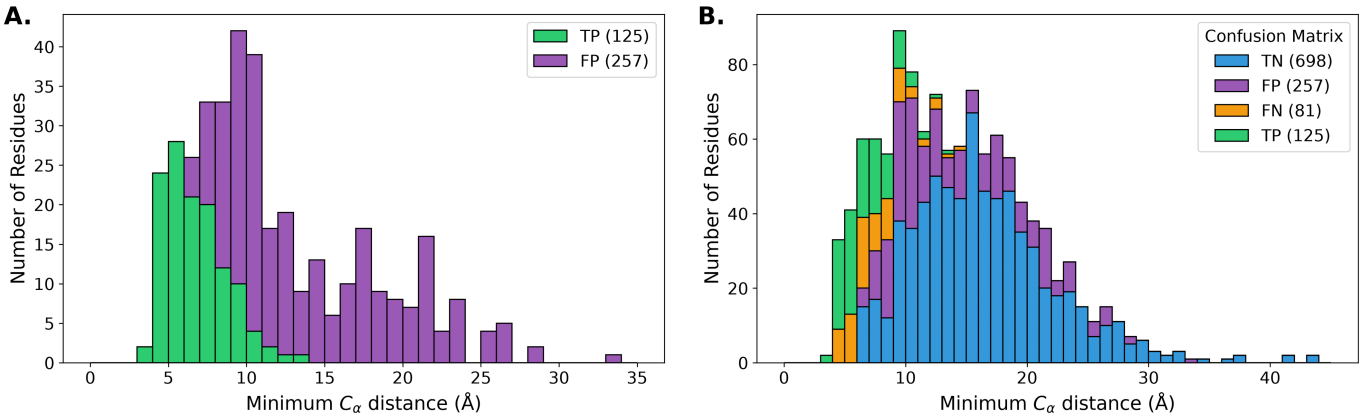
apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC	apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
	1YWI	6558	6559	0.78	0.72	7QCX	1D5G	34688	4516	0.27	0.26
	2MCN	16641	19447	0.79	0.71		2L4T	17254	17255	0.31	0.22
	5I22	4871	30010	0.56	0.50		2LOZ	10318	17364	0.31	0.19
	2ROZ	10235	10236	0.59	0.50		2FIN	6809	7024	0.19	0.19
	1N7T	18785	5631	0.45	0.40		2NMB	4683	4263	0.19	0.17
	1DDM	4683	4263	0.33	0.35		1VJ6	5131	6060	0.30	0.16
	2KPL	6911	16559	0.43	0.35		2HUG	6592	7241	0.38	0.14
1W1F	1WA7	6261	6456	0.48	0.32	2KQ3	2KHS	16585	15357	0.34	0.13
	2FFK	6809	7024	0.30	0.31		2KOH	27205	16507	0.26	0.06
	2M0U	18824	18825	0.35	0.31		2G35	15792	7061	0.17	0.04
	2L29	17128	17127	0.27	0.30		2KVM	15674	16778	0.33	-0.07
1SSF	2LVM	5878	18579	0.33	0.28		2JMF	6262	15016	0.00	-0.19



SI Fig. S4. Confusion Matrix Histograms for All Beta Receptors $(n = 24)$

SI Table S6. Data for Alpha and Beta (a+b) Receptors

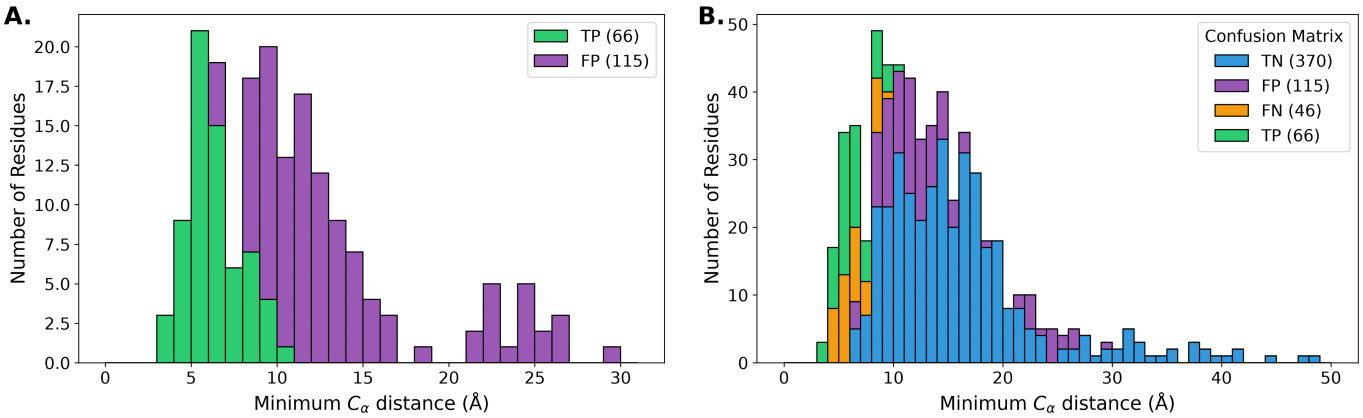
apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC	apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
	2M0G	18802	18808	0.48	0.52		2K6Q	5958	15877	0.37	0.26
	2PLD	5318	5310	0.59	0.52		6H8C	18827	34307	0.31	0.23
	2LI5	16835	17879	0.57	0.51		1KLQ	52275	5299	0.33	0.23
	2K7A	5461	16809	0.50	0.45		2MUR	25229	25230	0.57	0.23
	2M0G	19034	18808	0.44	0.33		2N55	52209	25694	0.24	-0.12
	2MUR	17769	25230	0.40	0.30		2K6D	52081	15866	0.08	-0.19
	5VF0	17769	30276	0.31	0.27						



SI Fig. S5. Confusion Matrix Histograms for Alpha and Beta (a+b) Receptors ($n = 13$)

SI Table S7. Data for BET-ET Domain Receptors

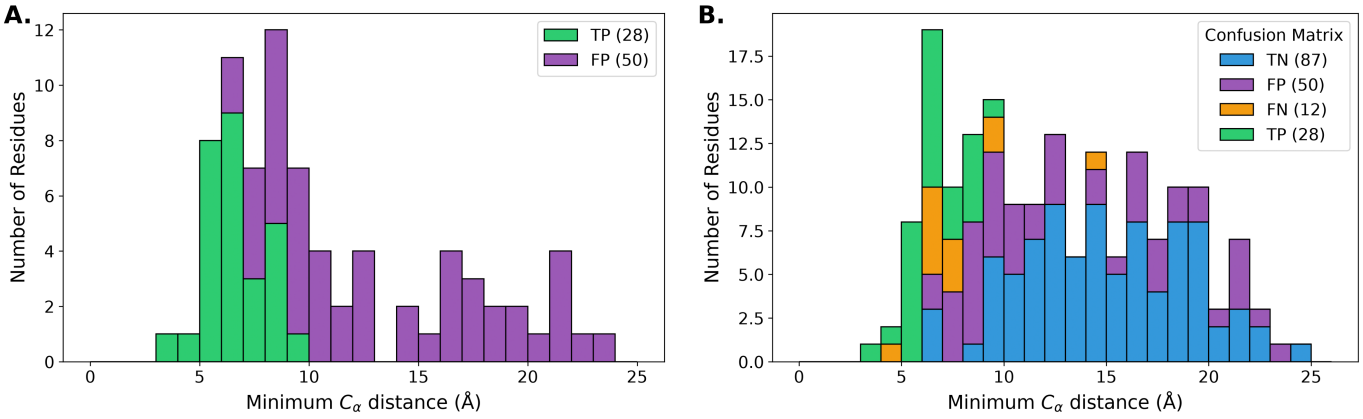
apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC	apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
7JMY	7JYZ	30782	30791	0.56	0.48	2JNS	2NCZ	15125	26041	0.46	0.35
2JNS	6BNH	15125	30373	0.53	0.42	2JNS	6BGG	30782	30367	0.32	0.29
7JMY	7JQ8	30782	30786	0.54	0.41	2JNS	2ND0	15125	26042	0.12	0.03
7JMY	7JYN	30782	30790	0.51	0.40	2JNS	2ND1	15125	26043	0.11	0.02



SI Fig. S6. Confusion Matrix Histograms for BET-ET Domain Receptors $(n = 8)$

SI Table S8. Data for Ubiquitin Domain Receptors

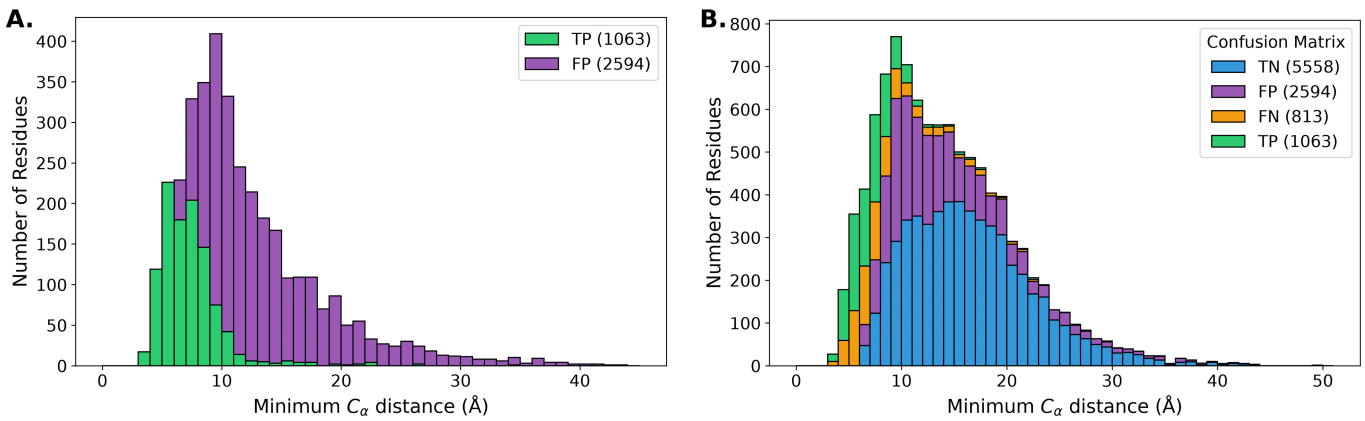
apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC	apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
	2MUR	17769	25230	0.40	0.30		2MUR	25229	25230	0.57	0.23
	5VF0	17769	30276	0.31	0.27						



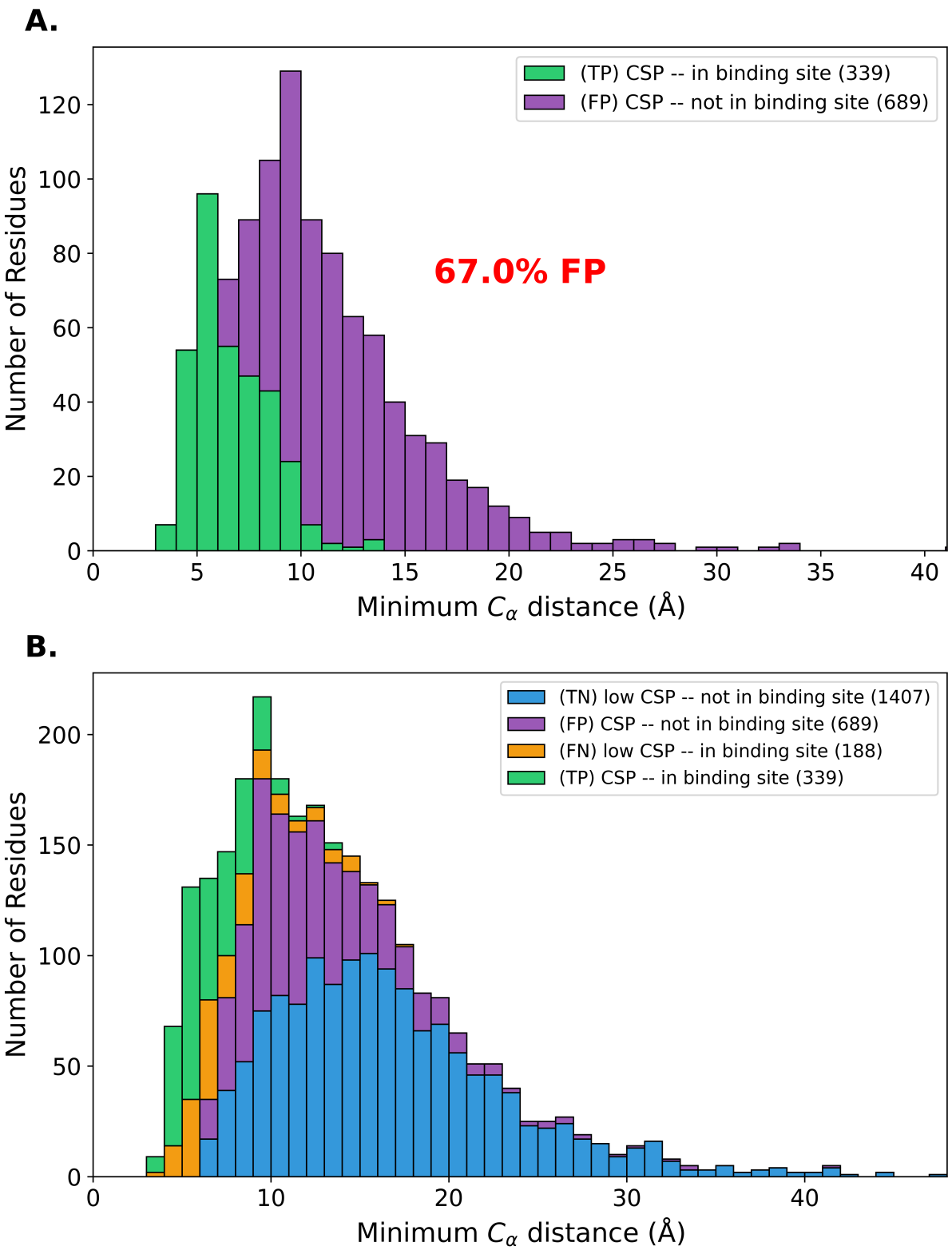
SI Fig. S7. Confusion Matrix Histograms for Ubiquitin Domain Receptors $(n = 3)$

SI Table S9. Data for targets with dissimilar apo/holo experimental conditions

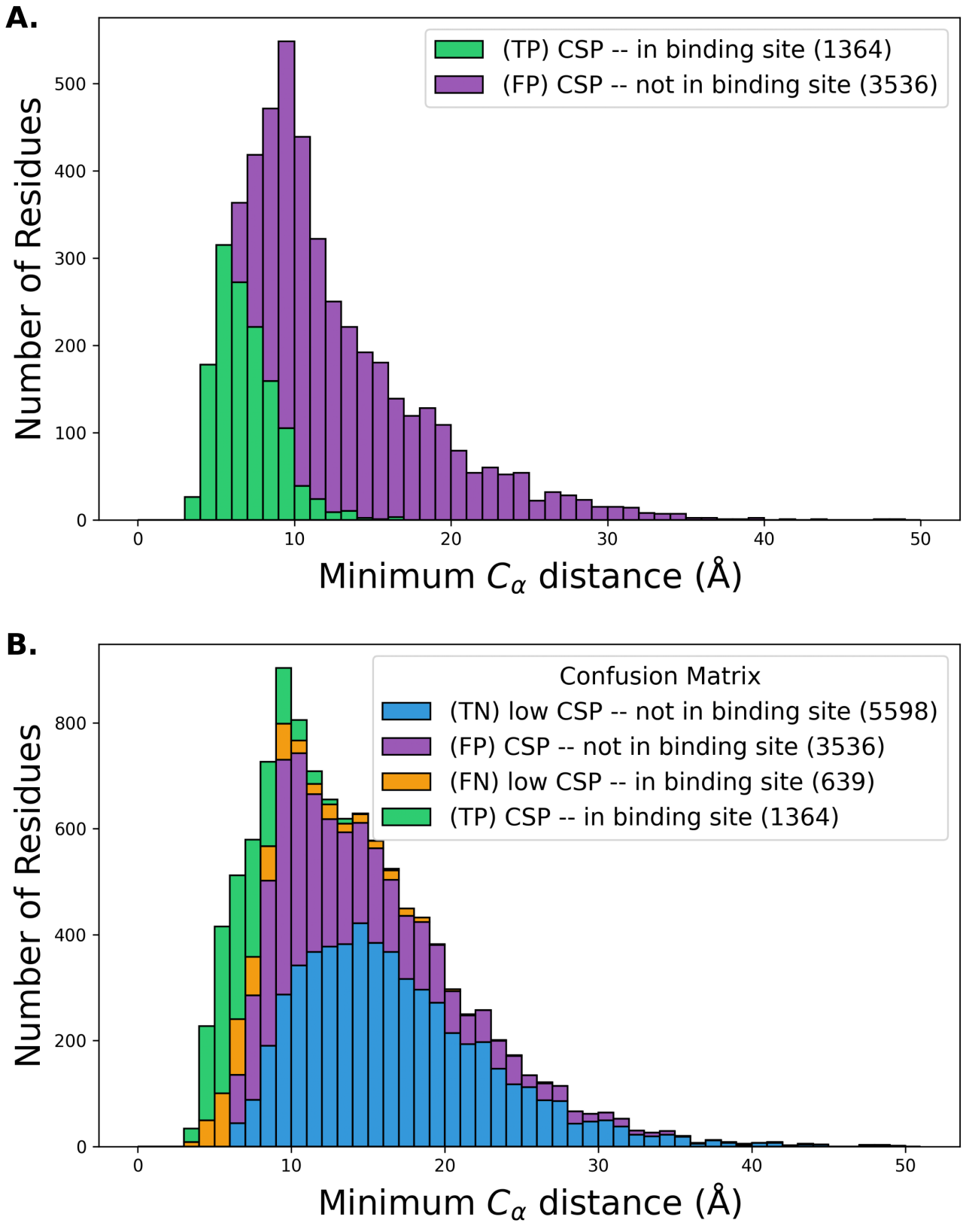
apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC	apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
	2MCN	6457	19447	0.76	0.67		1I5H	26698	4963	0.40	0.20
	6K5R	6801	36259	0.62	0.55		6FGP	16803	34232	0.38	0.20
	2LZ6	6457	18737	0.62	0.54	1N3H	2L1C	5566	17080	0.37	0.19
	2AFF	5959	6748	0.67	0.53	2GS0	5URN	6225	30243	0.27	0.19
	2MPM	4155	19989	0.59	0.50		2JQR	4566	15301	0.09	0.18
2GS0	2LOX	6225	18229	0.55	0.49	2L0I	5LVF	17044	34041	0.28	0.18
5AHT	2KPZ	25349	16574	0.61	0.48		2RSE	50325	11471	0.10	0.18
2GS0	2N0Y	6225	25540	0.50	0.47	2KJD	2M0V	16637	18826	0.21	0.17
	7ZEY	16983	34727	0.67	0.46		2LL7	51289	18028	0.33	0.17
1XWH	2KE1	6374	16510	0.68	0.45		5GOW	4901	36013	0.39	0.17
2AWT	2N9E	6801	25901	0.40	0.43		2MPS	6612	18876	0.33	0.17
	2G3Q	4769	7002	0.46	0.43		2L1R	30966	17103	0.35	0.15
2GS0	2M14	6225	18842	0.41	0.42	5YDY	2LAW	36115	17538	0.42	0.15
2GS0	2N23	6225	25584	0.40	0.42		5XV8	4901	36101	0.40	0.15
	2C52	15398	6874	0.56	0.41		2MV7	18094	19516	0.38	0.14
2RQT	2RQU	11081	11082	0.60	0.41		1LXF	30966	5386	0.37	0.14
	2KID	4879	16270	0.39	0.40		5JYV	30092	30093	0.31	0.13
2KM4	2L0I	16411	17044	0.41	0.39		2KNE	51289	16465	0.36	0.13
	7OVC	6546	34638	0.47	0.39	1V49	2LUE	5958	18518	0.23	0.12
	2MRE	6457	25070	0.45	0.39		2H7D	15792	7150	0.25	0.12
	2KWU	17769	16880	0.42	0.39		2KA6	4789	16015	0.52	0.11
2LSG	2LSJ	18431	18433	0.36	0.37		2LL6	51289	18027	0.32	0.11
2LD9	2MBH	15410	19399	0.43	0.36		2M86	15945	19230	0.25	0.10
2LPC	2M04	18250	18793	0.47	0.35		2MGU	51289	19604	0.42	0.09
2JQ6	2KFH	15279	16181	0.42	0.35		2MZP	30966	25495	0.18	0.07
	6K5T	25299	36260	0.48	0.35		2KJE	4789	16318	0.48	0.07
	2LTO	10276	18490	0.27	0.34	2M03	6IJQ	18792	36133	0.13	0.06
	2M5A	5656	19043	0.53	0.33		2L53	51289	17264	0.28	0.06
	2MS4	2208	25104	0.33	0.33		2KFT	6374	16878	0.24	0.05
	1CF4	18251	4700	0.52	0.32		2MMA	19849	19850	0.18	0.05
	2KWV	17769	16885	0.38	0.32	2E5E	2L7U	7364	17378	0.15	0.05
	1M4P	50765	5532	0.29	0.32		2MZD	34231	25484	0.34	0.04
	2KVQ	15490	16788	0.27	0.31		6BUT	51289	27095	0.25	0.03
5AHT	2KQ0	25349	16575	0.52	0.31		2RMK	5511	11010	0.18	0.03
	2KJ4	4276	16311	0.35	0.30		7F7X	17769	36427	0.19	0.03
	2RUK	4901	11578	0.44	0.30		2RS9	19125	11463	0.11	0.03
	2N4Q	18094	25677	0.40	0.30		2RSY	7141	11508	0.28	0.03
1U2N	1L8C	6268	5327	0.63	0.30		6XMN	30316	50331	0.37	0.02
	2LY4	11532	18709	0.40	0.30		6B1G	27072	30345	0.27	0.02
	2MBB	15410	19394	0.45	0.29		2N1A	6304	25553	0.07	0.01
2GS0	2MKR	6225	19791	0.30	0.29		5J7J	51289	30062	0.34	0.01
	9C5E	6457	31177	0.33	0.28		2MKP	25495	19789	0.24	0.01
	6CO4	30824	25979	0.33	0.28		5TP6	51289	30196	0.23	-0.01
	7ZEY	34724	34727	0.39	0.28		2N8J	51289	25852	0.19	-0.01
	2MG5	51289	19586	0.44	0.27	5YDX	2LAY	36114	17540	0.30	-0.02
	2MZW	17503	25504	0.38	0.27		2M55	51289	19036	0.21	-0.03
	2M3M	17373	17942	0.44	0.26		2N83	25883	25833	0.28	-0.05
	2KNH	7396	16467	0.27	0.25		2LV6	51289	18556	0.36	-0.06
2L0I	5M9D	17044	34058	0.35	0.25		2LAS	50128	25553	0.11	-0.07
	2RVB	4901	11594	0.45	0.23		2NBV	30824	25995	0.11	-0.07
	5IAY	30703	30019	0.23	0.23		1NPQ	4232	5738	0.23	-0.07
	7L8V	26682	27692	0.30	0.23		2MNU	4283	19906	0.15	-0.09
1U2N	2KA4	6268	16014	0.58	0.22		7NQC	51289	34608	0.14	-0.11
	2MH0	34231	19610	0.35	0.21		2MWN	15792	25343	0.00	-0.13



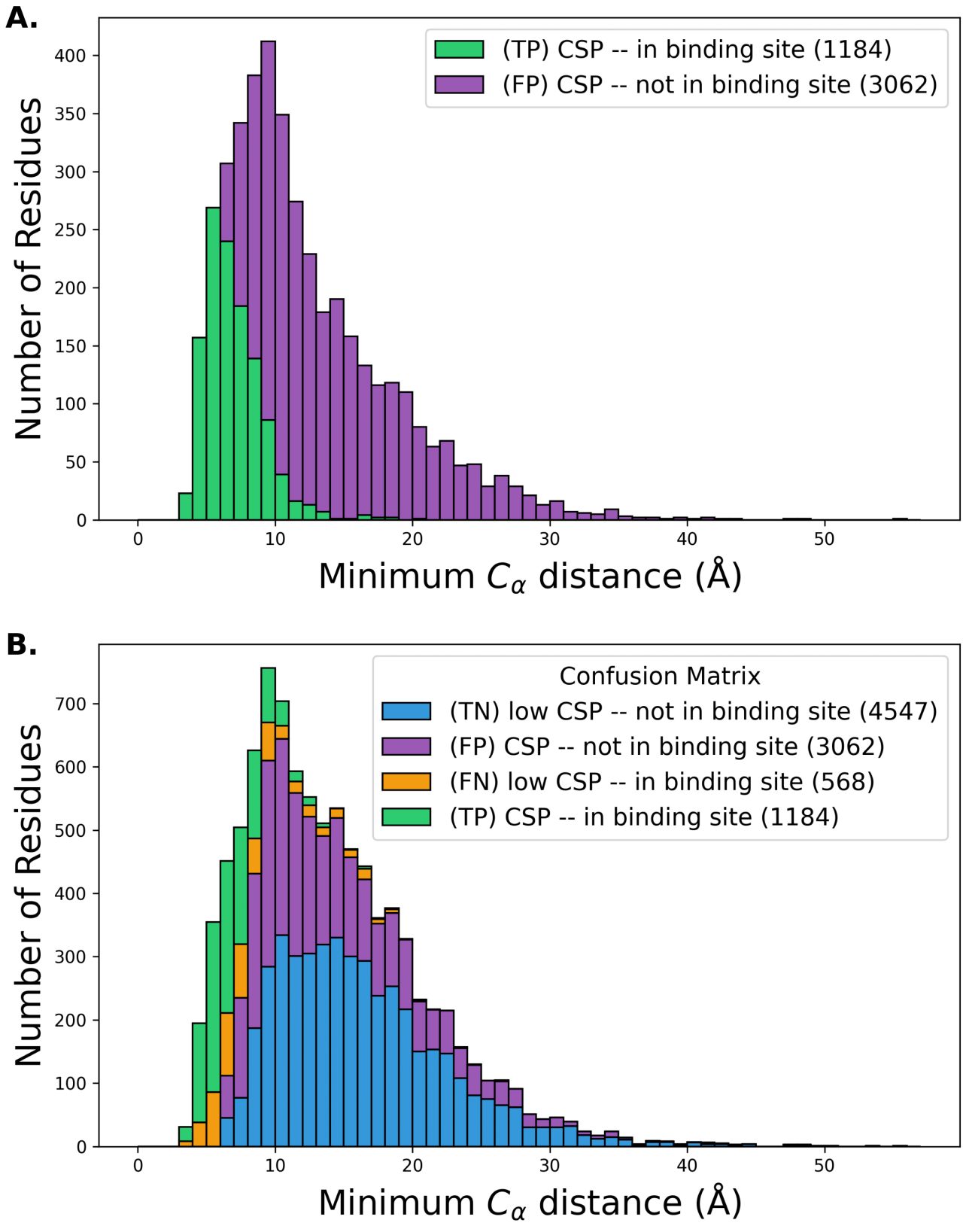
SI Fig. S8. Confusion Matrix Histograms for targets with dissimilar apo/holo experimental conditions ($n = 108$). The $n = 108$ subset for SI Fig. S8 and SI Table S9 come from the subset of targets from the complete $n = 243$ CSPdb with a pH difference between apo and holo experiments > 0.5 units or temperature difference > 5°C.



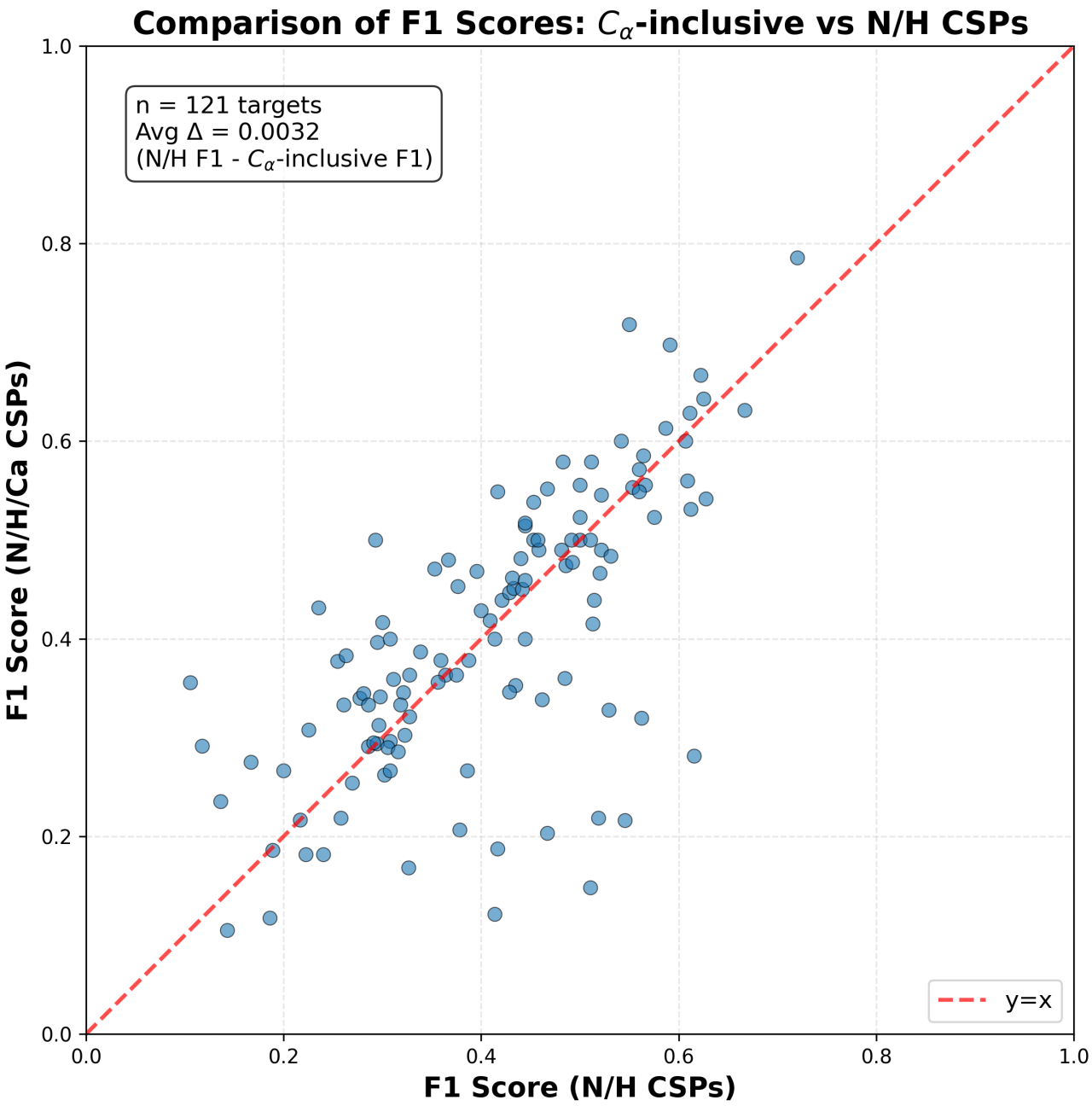
SI Fig. S9. Figure 3 histograms for the same-author/same-study subset. Targets are the curated pairs in `CSP_UBQ_ph0.5_temp5C_same_author_list.csv` ($n = 32$ with pipeline outputs of 33 curated pairs): buffer-similar apo/holo systems that share a study or last author between apo and holo chemical-shift depositions. **A.** significant-CSP TP vs FP by minimum Cα distance to the interface; **B.** full confusion-matrix stacks (TN/FP/FN/TP). This figure reports results on a very high-fidelity subset, and the results are very similar to those reported in Figure 3: 67.0% of CSPs are distal vs 69.4% of CSPs are distal in the $n = 135$ subset reported in Figure 3.



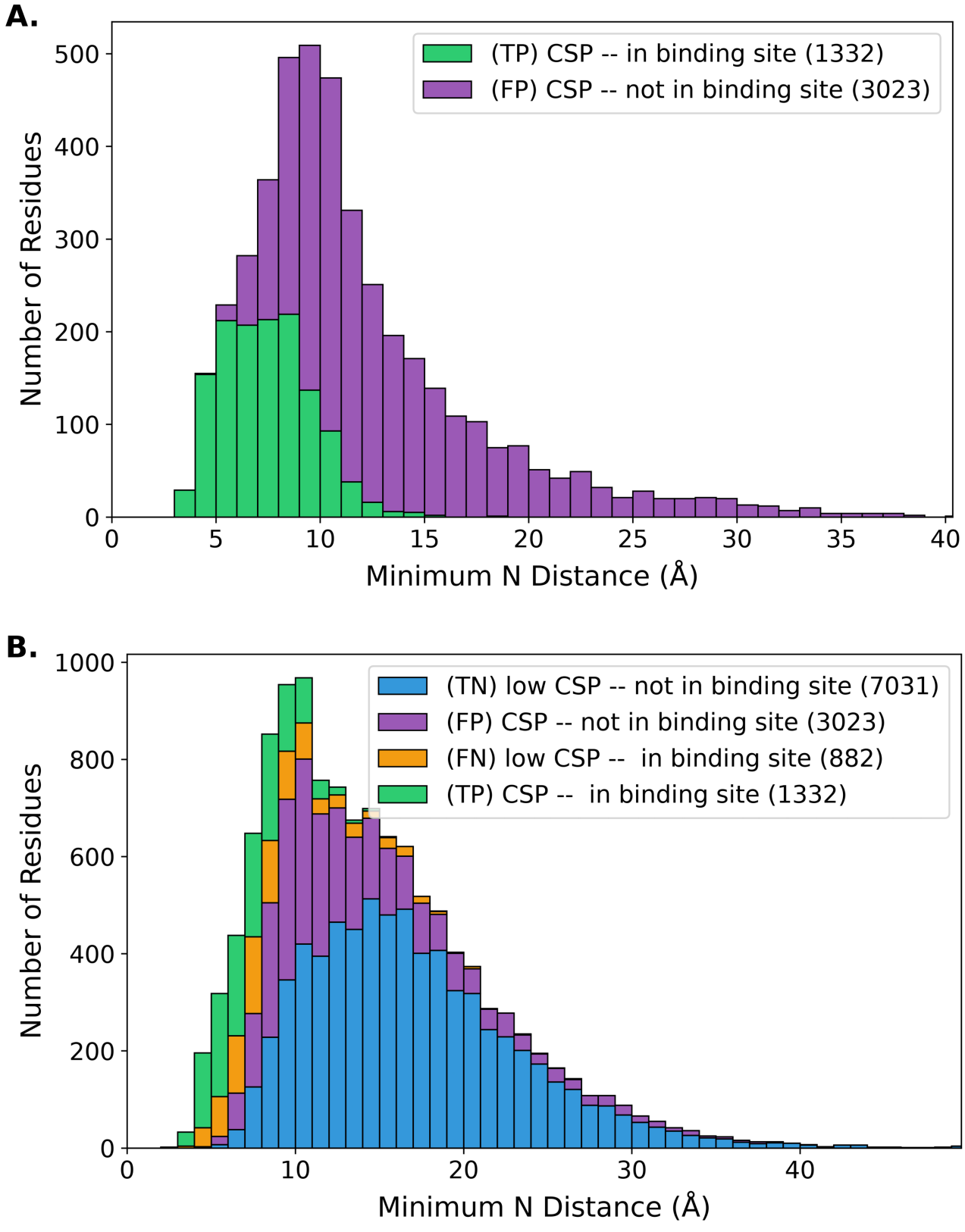
SI Fig. S10. Confusion histograms calculated with Eqn. 2 for the $n = 121$ buffer-similar systems from `CSP_UBQ_ph0.5_temp5C.csv` with CA row coverage $> 50\%$. These results are very similar to CSPs calculated using SI Equation S1. 72.2% of CSPs using SI Equation S2 are distant from the binding site compared to 69.4% using SI Equation S1.



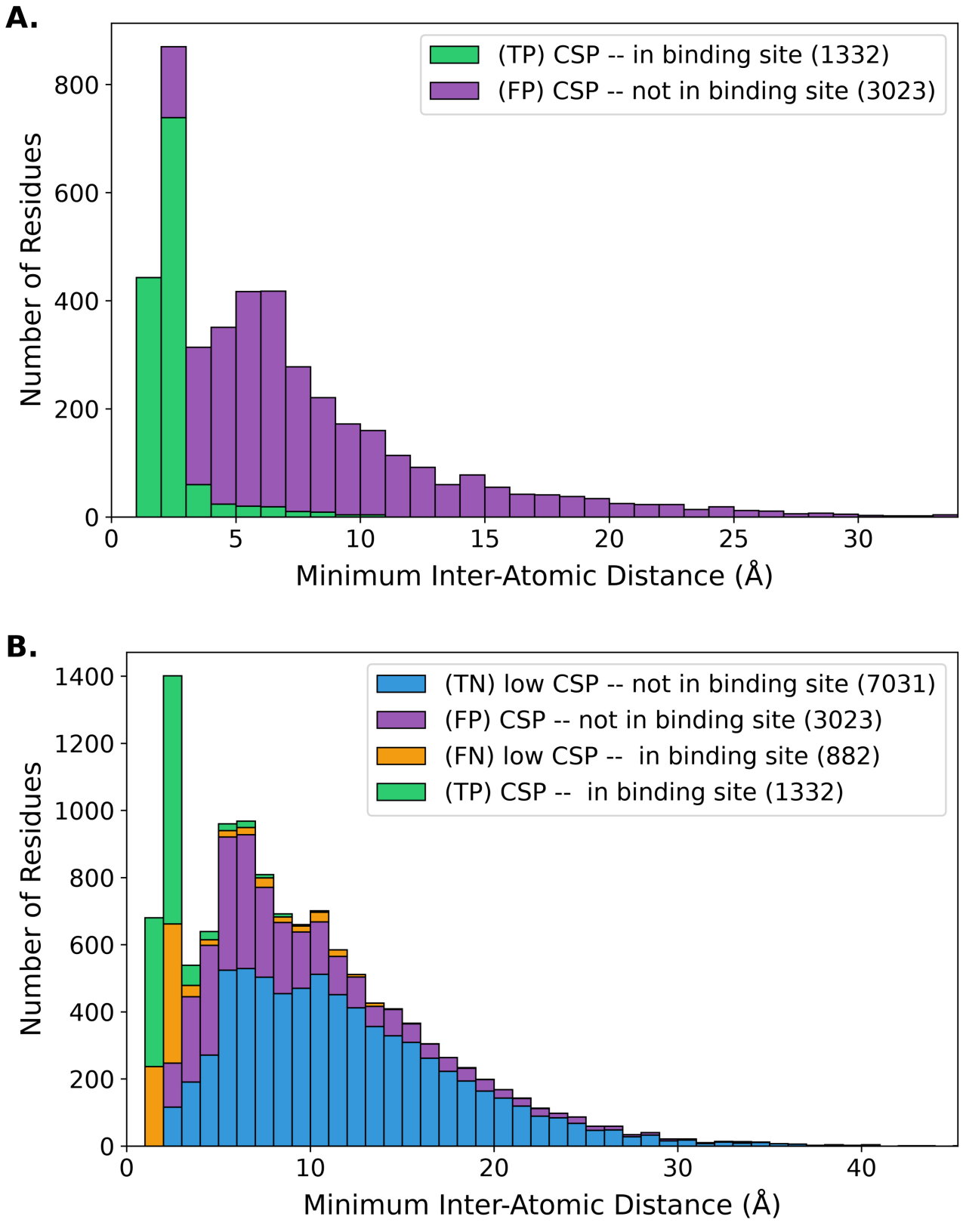
SI Fig. S11. Confusion histograms calculated with Eqn. 3 for the buffer-similar systems from `CSP_UBQ_ph0.5_temp5C.csv` that also have Hα and Cα shifts in both apo and holo experiments ($n = 101$). These results are very similar to CSPs calculated using SI Equation S1. 72.2% of CSPs using SI Equation S3 are distant from the binding site compared to 69.4% using SI Equation S1.



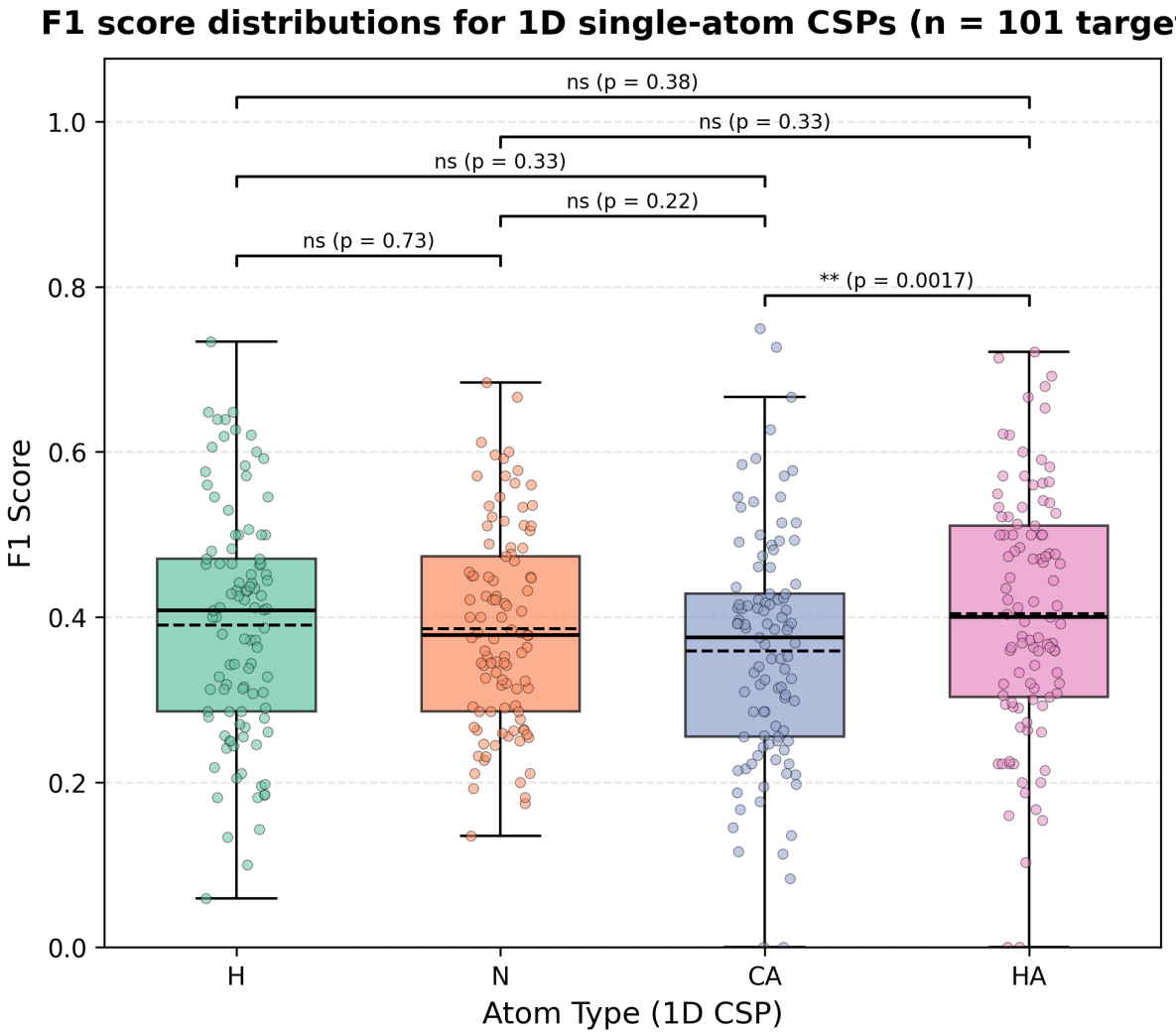
SI Fig. S12. F1 for Eqn. 1 CSPs vs Eqn. 2 CSPs ($n = 121$ buffer-similar systems from `CSP_UBQ_ph0.5_temp5C.csv` with CA row coverage $> 50\%$). F1 scores on the y axis use SI Equation S2 (combined HN+Cα); F1 scores on the x axis use SI Equation S1 (combined HN). The F1 scores here demonstrate that per-target there are very few differences in the distribution of CSPs distant from the binding site across the $n = 121$ subset of targets with CA CSPs and similar experimental conditions.



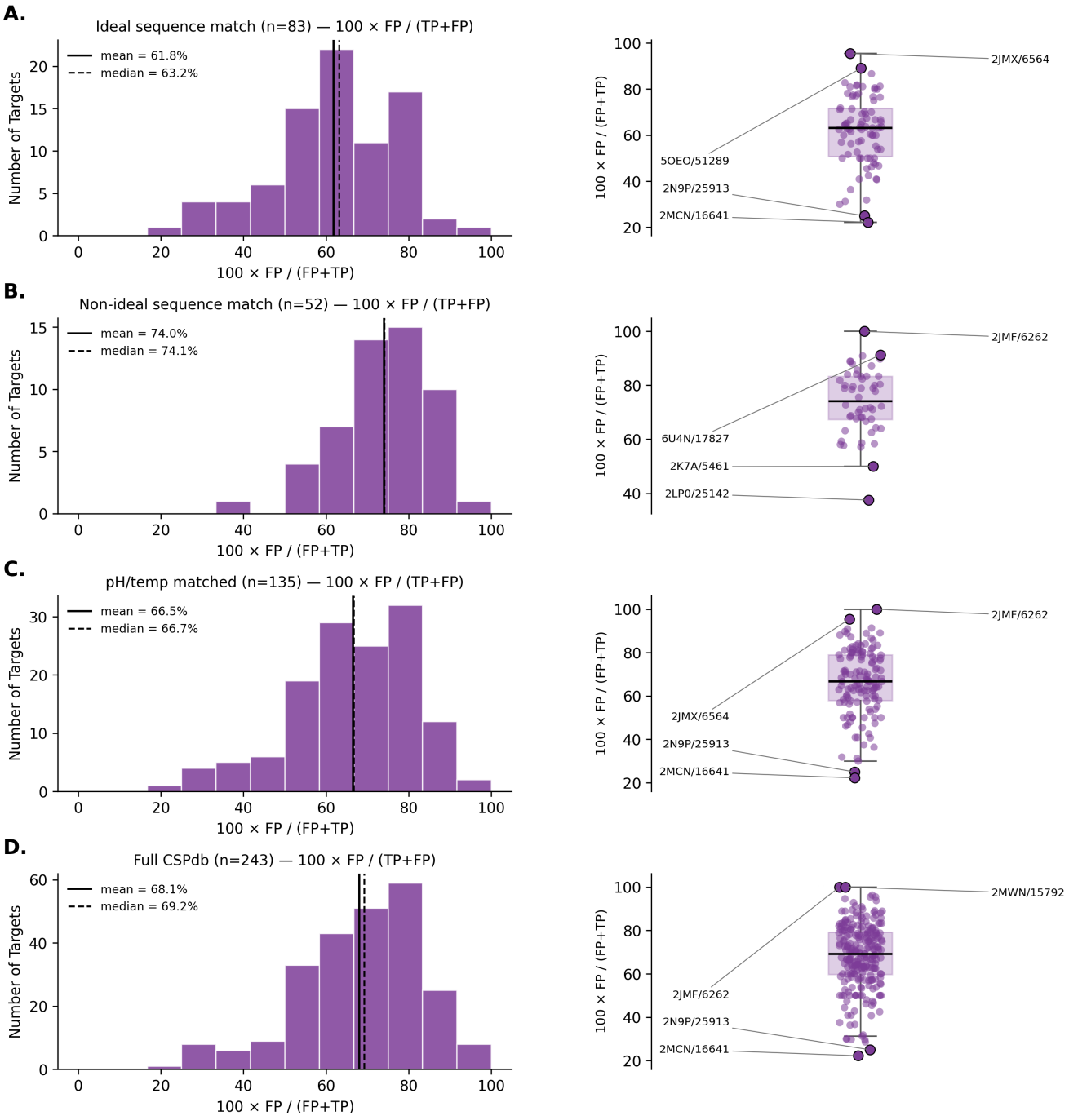
SI Fig. S13. Confusion histograms by closest interchain N–N distance (n = 135 buffer-similar systems from `CSP_UBQ_ph0.5_temp5C.csv` with pipeline outputs).



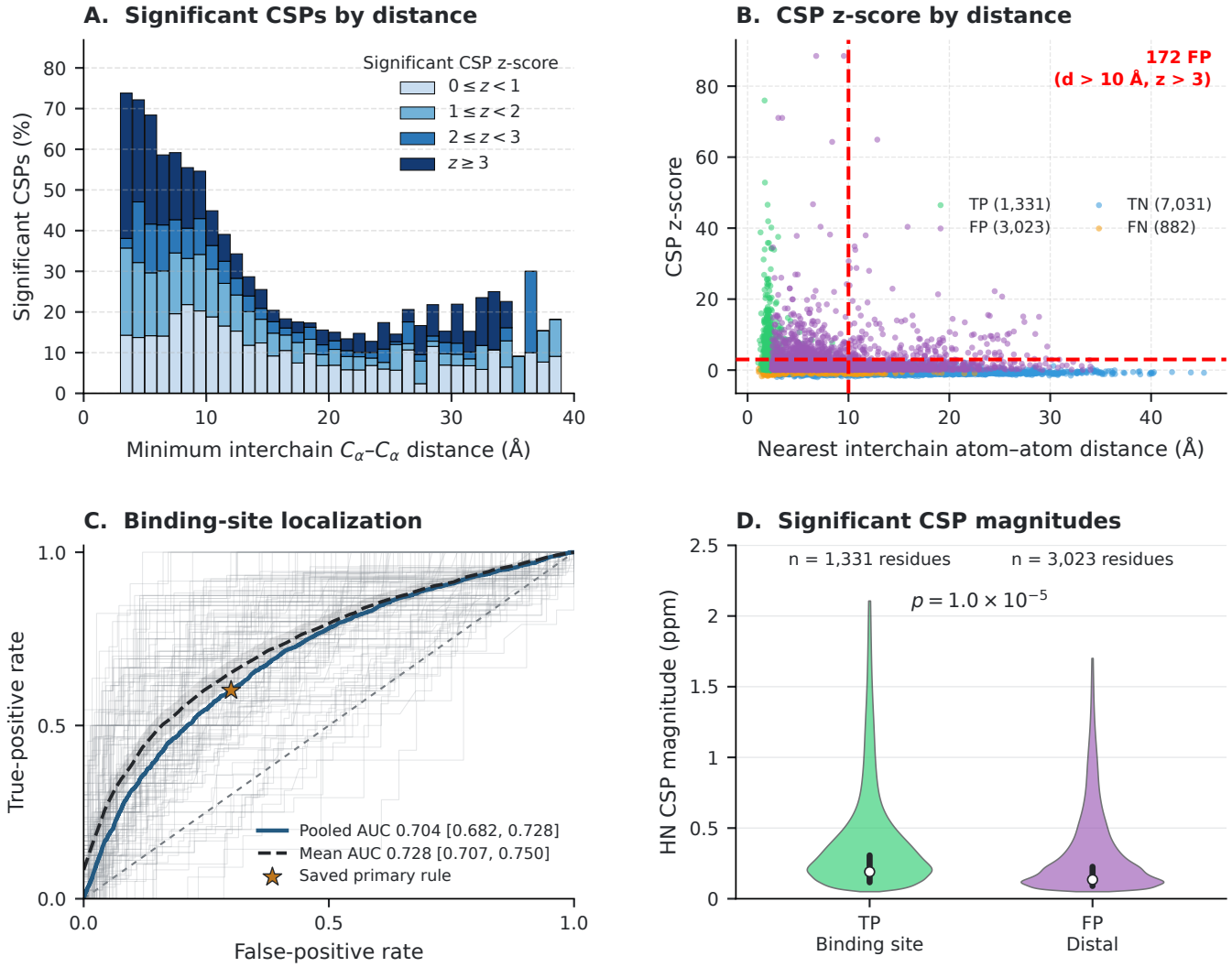
SI Fig. S14. Confusion histograms by closest interchain any-atom distance ($n = 135$ buffer-similar systems from `CSP_UBQ_ph0.5_temp5C.csv` with pipeline outputs).



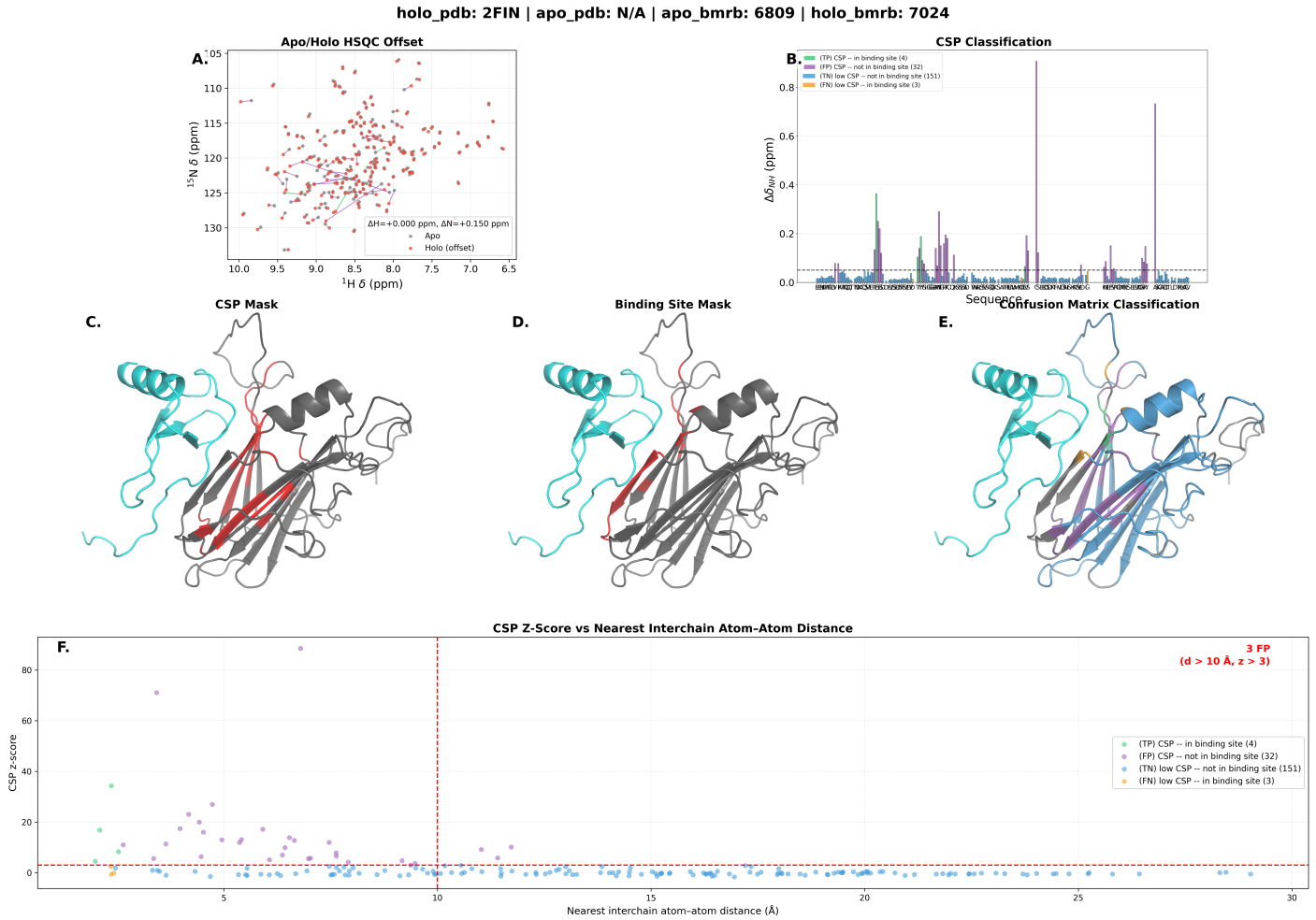
SI Fig. S15. Per-target F1 scores for 1D H/N/Cα/Hα CSPs ($n = 101$ buffer-similar systems from `CSP_UBQ_ph0.5_temp5C.csv` with paired H/N/CA/HA F1 scores). Boxes show the per-target F1 distribution for 1D CSPs of each nucleus. Pairwise comparisons are two-sided Wilcoxon signed-rank tests for all unordered atom pairs, with Holm–Bonferroni adjustment of p-values. After Holm–Bonferroni adjustment, H, N, and Cα F1 scores do not differ (all adjusted $p > 0.05$). Cα versus Hα is the exception (Holm-adjusted $p = 0.0017$): median per-target F1 is 0.38 for Cα and 0.40 for Hα. Distal CSPs still occur for every nucleus at a similar frequency. Apo/holo offsets for 1D CSPs are chosen independently for each nucleus by a 1D grid search on unreferenced shifts, using the same bounds as the joint searches (^{1}H$_{\mathrm{N}}$ $\pm$0.2 ppm, step 0.01 ppm; ^{15}N $\pm$1.5 ppm, step 0.05 ppm; ^{13}Cα $\pm$2 ppm, step 0.05 ppm; ^{1}Hα $\pm$0.12 ppm, step 0.01 ppm). The CSPs calculated to retrieve an F1 score per target are calculated as $(\mathrm{dd}\langle\mathrm{Shift}\rangle^2)^{1/2}$ and the threshold for significance is set as the max of the cleaned mean and 0.05 ppm, for all 1D CSPs shown in this figure.



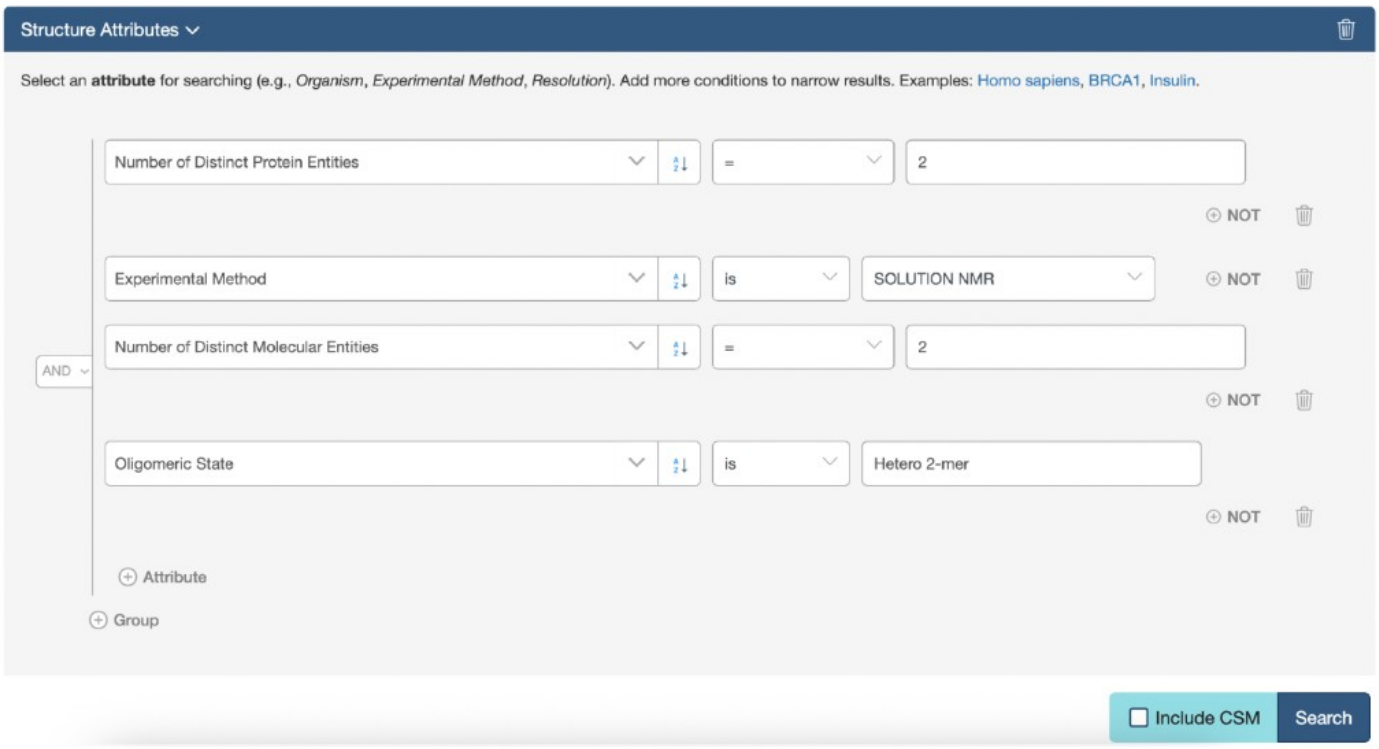
SI Fig. S16. Per-target false-positive rates under the primary CSP cutoff max(cleaned mean, 0.05 ppm). **A.** ideal sequence match ($n = 83$); **B.** non-ideal sequence match ($n = 52$); **C.** pH/temp matched ($n = 135$); **D.** full CSPdb ($n = 243$). For each cohort, panels report $100 \times \mathrm{FP}/(\mathrm{TP} + \mathrm{FP})$ among significant CSPs. This figure shows per-target false positive rates across four subsets, and reveals there are minimal differences between these four subsets.



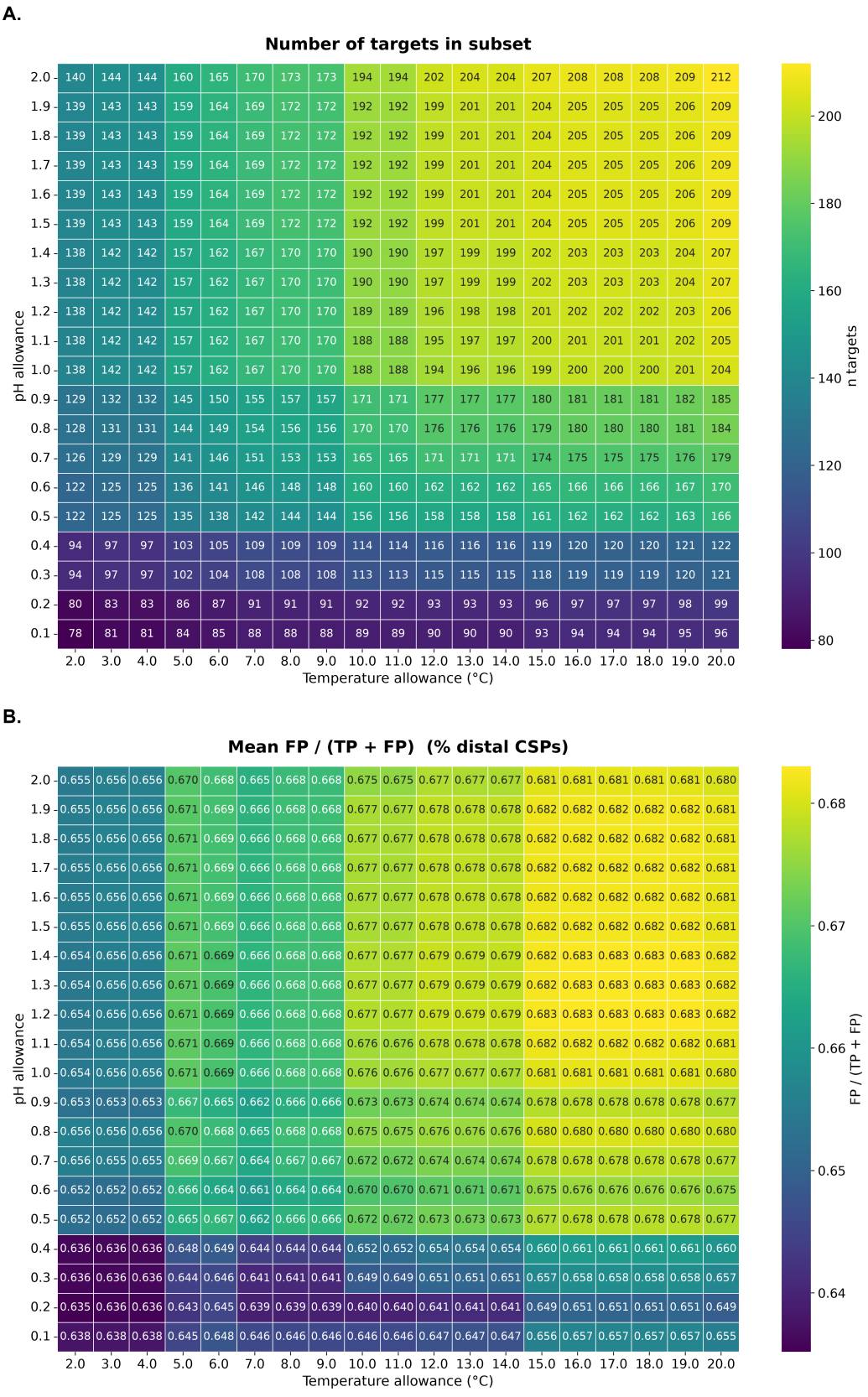
SI Fig. S17. CSPs and structural binding-site localization. All panels use 135 pH- and temperature-matched apo/holo targets. The per-target cutoff is max(cleaned mean HN CSP, 0.05 ppm). **A.** Percentage of residues with a significant HN CSP among those with a recorded CSP in each 1 Å bin of minimum interchain C_α–C_α distance. Stacks partition the significant fraction by CSP z-score. **B.** Per-residue CSP z-score versus nearest interchain atom–atom distance. For residue i, $z_i = (\mathrm{CSP}_i - \mu_{\mathrm{clean}})/\sigma_{\mathrm{clean}}$, where μ_{clean} and σ_{clean} are the mean and population standard deviation of the target's HN CSP list after iterative removal of high outliers. Colors indicate true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN) relative to the structural binding-site definition. Dashed lines mark $z = 3$ and $d = 10$ Å. The 172 annotated FPs exceed both thresholds. **C.** Receiver operating characteristic (ROC) curves across HN CSP thresholds for 12,267 eligible residues. Gray curves show individual targets, blue shows the pooled ROC using CSP divided by the target-specific cutoff, and the black dashed curve shows the equal-target mean ROC. The pooled area under the curve (AUC) is 0.704 (95% CI 0.682–0.728), and the mean per-target AUC is 0.728 (95% CI 0.707–0.750). The band shows the pointwise 95% interval for the mean ROC. Intervals use 2,000 whole-target bootstrap replicates. The gray diagonal indicates chance, and the star marks the pooled operating point from the saved primary significance calls. **D.** HN CSP magnitudes for 1,331 significant binding-site residues (TP, green) and 3,023 significant distal residues (FP, purple) under the saved primary rule, pooled across targets. Scott-bandwidth kernel densities are estimated and normalized to equal area in $\log_{10}$ CSP space and bounded by the observed ranges. White dots mark medians and black bars mark interquartile ranges. The y-axis is linear (ppm). The symmetric Kullback–Leibler divergence was 0.0941 nats (20 shared log-spaced bins, 0.5 pseudocount per bin), with $p = 1.0 \times 10^{-5}$ from 99,999 within-target label permutations.



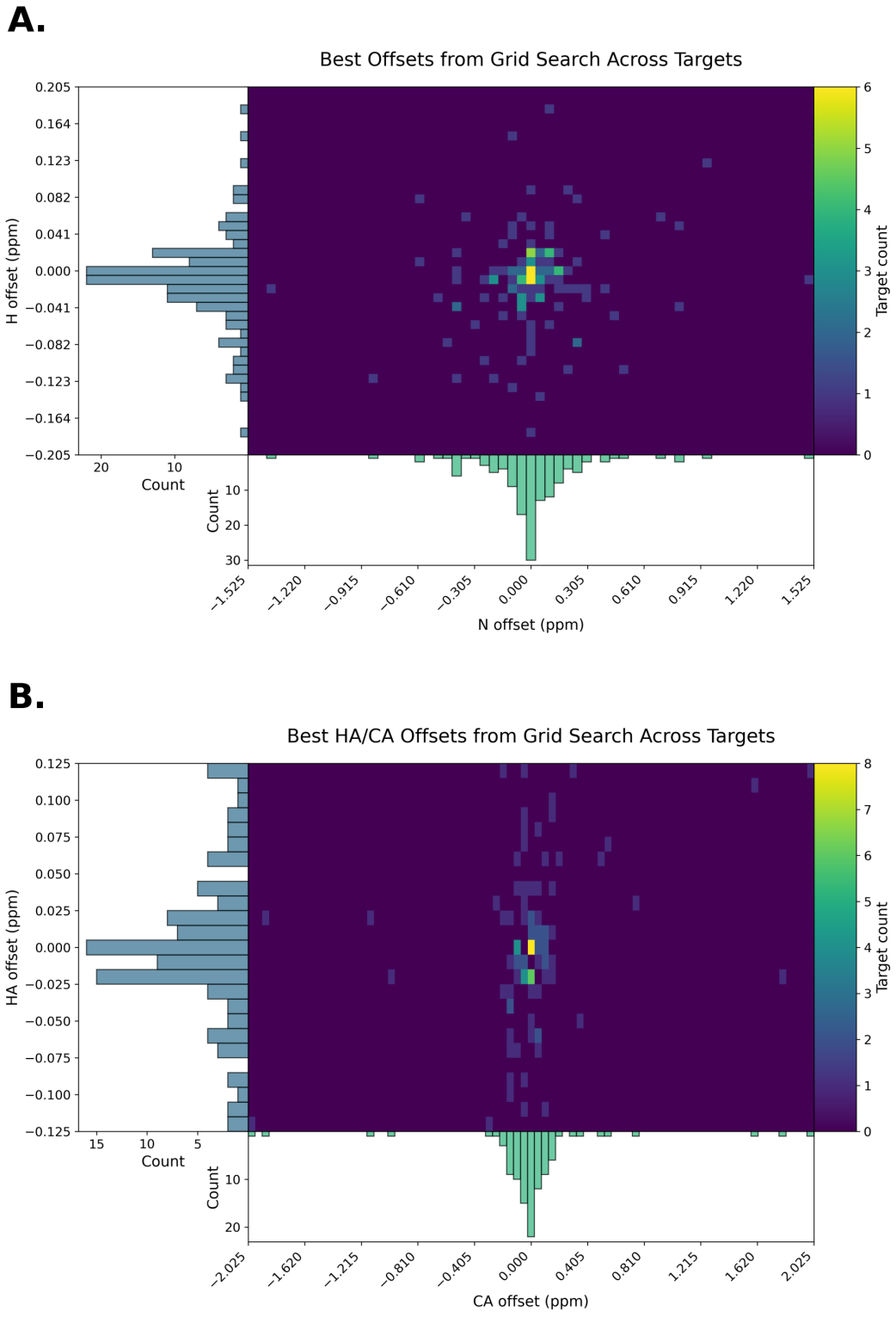
SI Fig. S18. Case study for holo 2FIN / apo BMRB 6809 (holo BMRB 7024). **A.** ^{15}N/^{1}H HSQC scatterplot overlaying the apo and holo assigned spectra with the optimal N/H offset applied. **B.** Calculated CSPs for receptor residues along the sequence, bars colored by confusion matrix classification. **C.** The mask of significant CSPs. **D.** The mask of binding-site residues. **E.** The confusion matrix classification projected onto the structure: distal CSPs are purple (FP), CSPs in the binding site are green (TP), non-CSPs in the binding site are orange (FN), and non-CSPs outside of the binding site are blue (TN). Residues without CSP measurements are shown in gray; ligands are colored cyan. **F.** CSP z versus nearest interchain atom–atom distance. Dashed lines mark $z = 3$ and $d = 10$ Å; the annotation counts false positives with $d > 10$ Å and $z > 3$. This plot shows monotonic decay of CSPs with increasing nearest interchain atom–atom distance. Residues in a set of β sheets further away from the ligand with significant shifts have lower CSPs than those closer to the binding site.



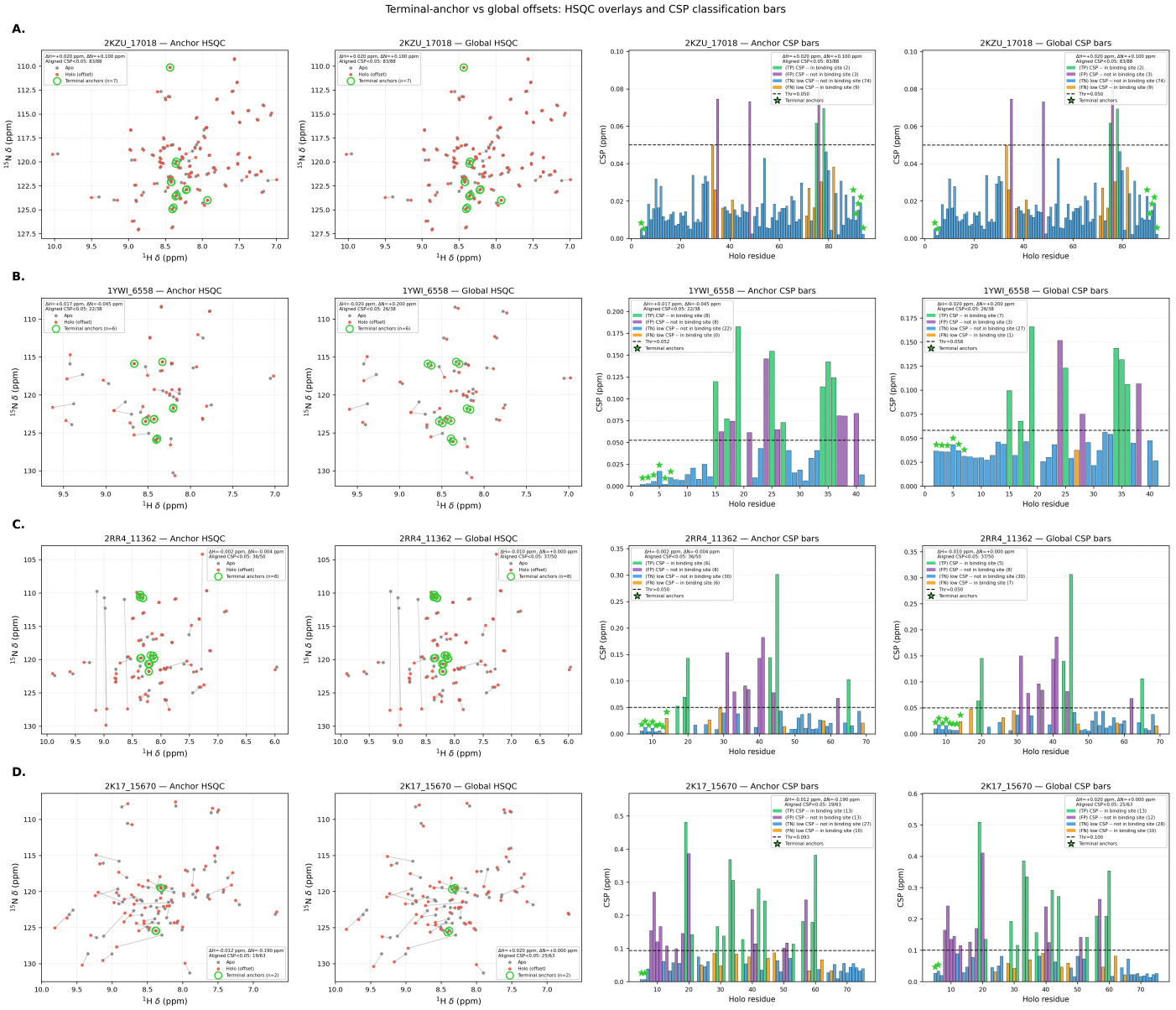
SI Fig. S19. PDB Advanced Search Settings. For the subset of targets manually scraped from the PDB, we restrict our analysis to these settings: entries resolved by Solution NMR, with two and only two distinct protein chains. At the time of curation, this advanced search resulted in 720 hits on the RCSB PDB. From this list of 720, entries were included in the CSP DB if there were N and H chemical shifts deposited in the BMRB for an identical sequence which was evaluated in both apo and holo experiments.



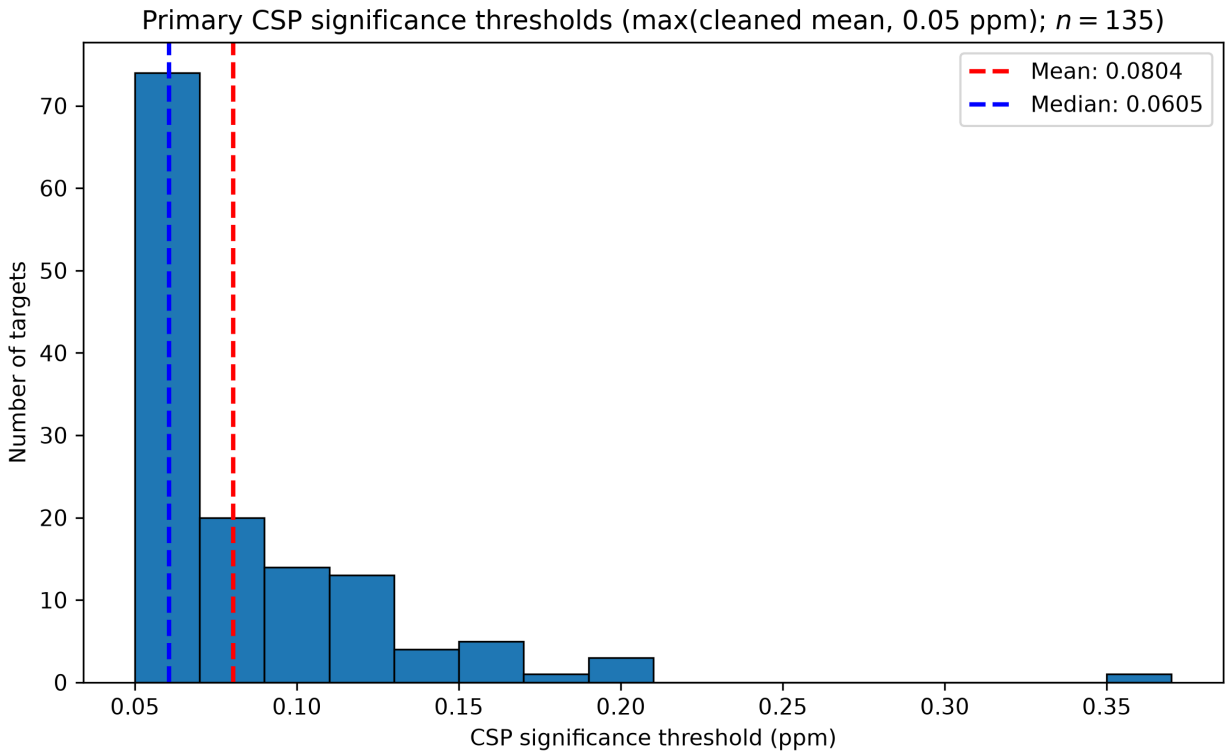
SI Fig. S20. Buffer pH/temperature threshold sweep. A. Number of apo/holo pairs retained as $|\Delta\text{pH}|$ and $|\Delta T|$ thresholds are tightened. **B.** Mean FP/(TP+FP) (percent distal CSPs) in the same subsets. This figure reveals the mean per-target false positive rate changes only slightly across all thresholds in apo/holo experimental condition differences from a minimum of 63.6% to a maximum of 68.3%.



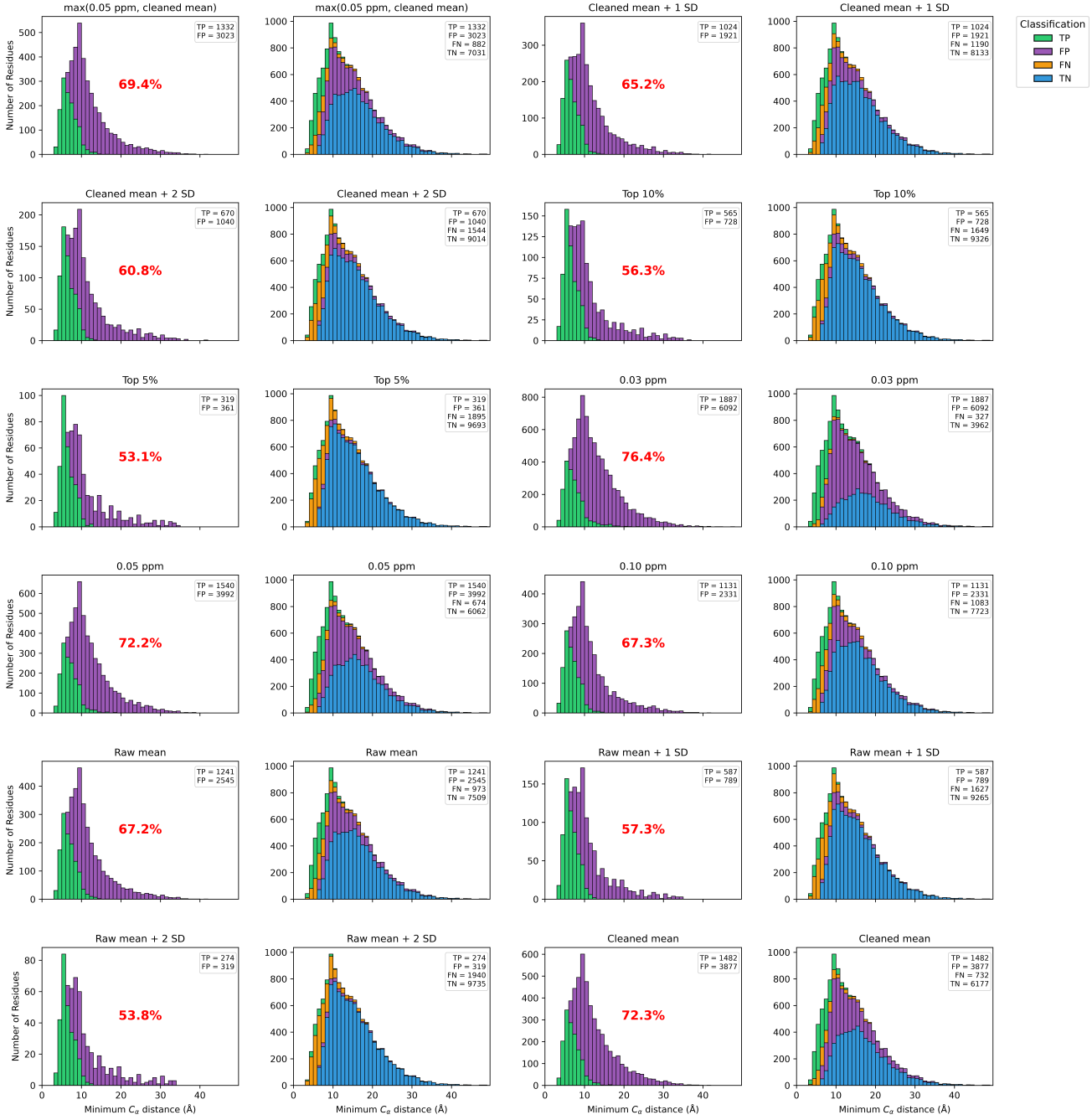
SI Fig. S21. Ideal chemical-shift offset grids for buffer-similar systems in `CSP_UBQ_ph0.5_temp5C.csv` with pipeline outputs. **A.** H/N referencing ($n = 135$): a grid search selects the H/N offsets that maximize aligned HN peak counts under the pipeline CSP criterion; the heatmap aggregates those optimal ($\Delta\delta_N$, $\Delta\delta_H$) pairs (color = number of targets). Marginal histograms show the one-dimensional distributions of best H (left) and N (bottom) offsets. **B.** HA/CA referencing ($n = 101$ systems with HA/CA grid-search CSVs): the same layout for offsets that maximize aligned HA/CA peak counts, with HA (left) and CA (bottom) marginals. Both nuclei pairs are peaked near zero. On the Cα window of ± 2 ppm, 87 of 101 Cα offsets lie within 0.2 ppm of zero and 5 have $|\text{offset}| \geq 1.5$ ppm, so most apo/holo pairs still require only small referencing corrections.



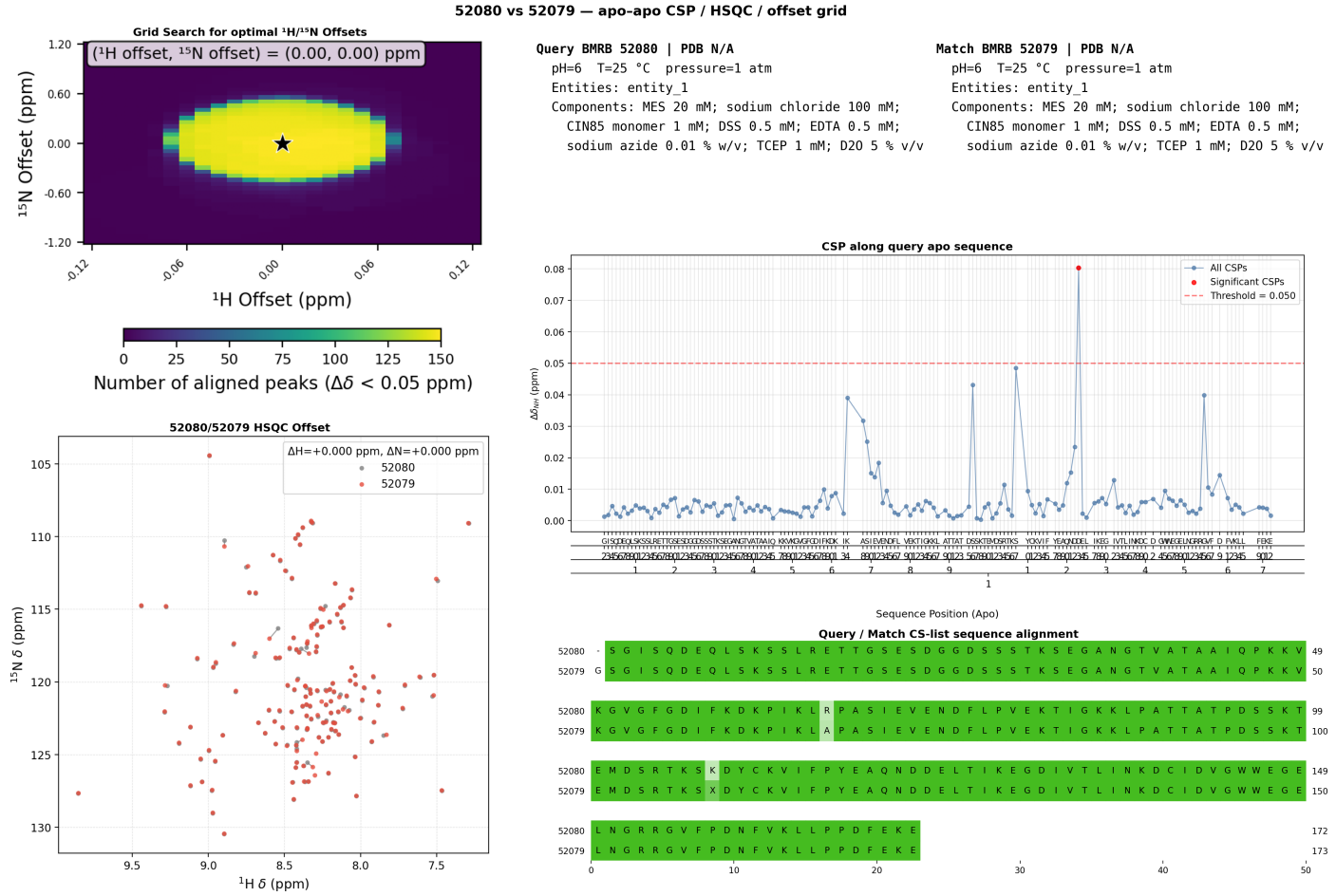
SI Fig. S22. Terminal-anchor vs global offsets: HSQC overlays and CSP classification bars for four targets (2KZU/17018, 1YWI/6658, 2RR4/11361, 2K17/15670). For each target (rows A.–D.): left pair shows apo (grey) vs offset holo (red) HSQC overlays using terminal-anchor RMSD offsets vs the global CSP-count grid; right pair shows corresponding CSP classification bars (TP/FP/TN/FN) under the primary significance threshold. Green markers denote terminal-anchor residues. This figure reveals an alternative approach to referencing apo and holo shift lists would be an appropriate solution to the problem, but is not generalizeable across the dataset. These results further demonstrate our approach of global optimization described in the Methods is an appropriate generalizeable solution to this problem.



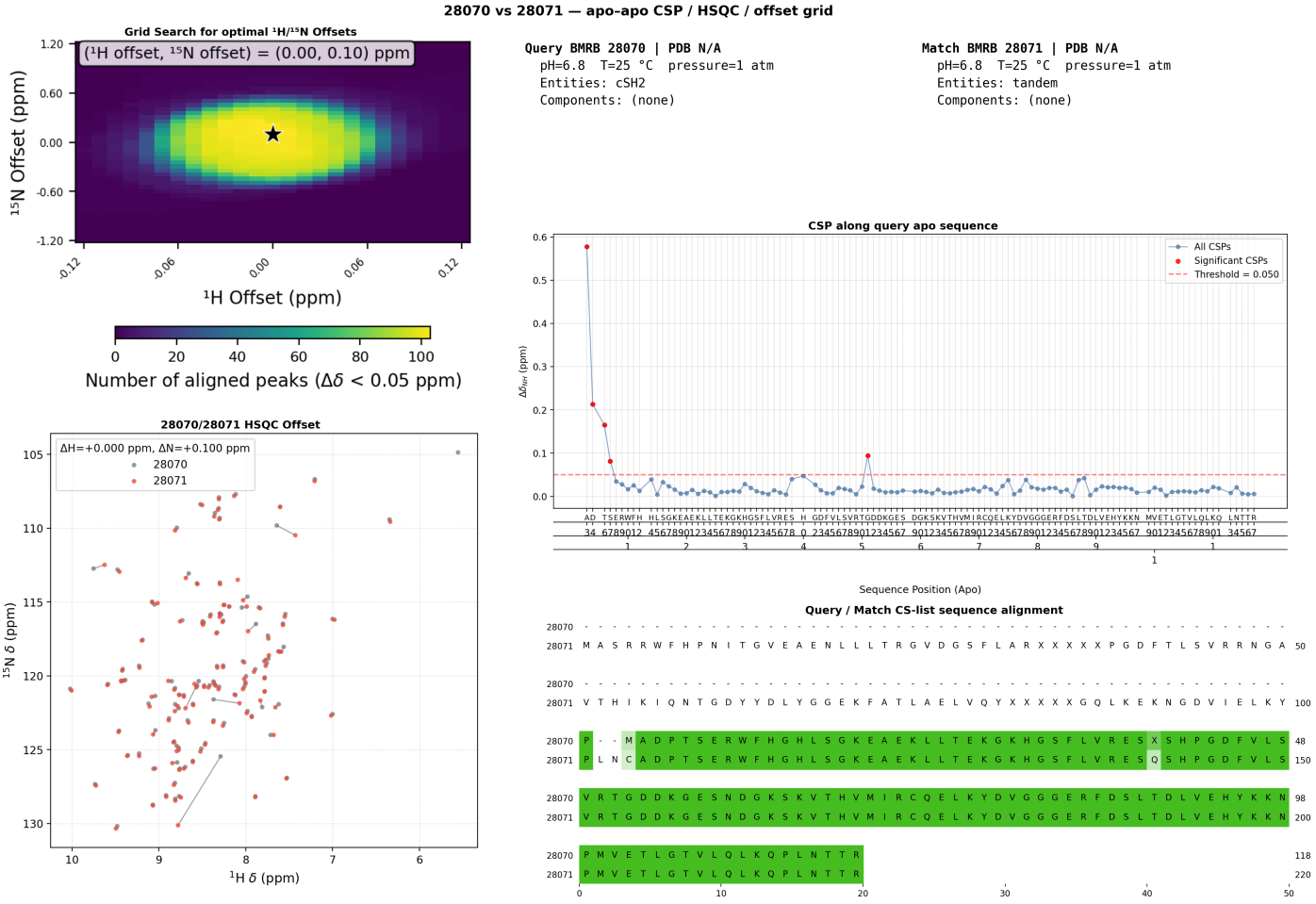
SI Fig. S23. Distribution of CSP significance thresholds (primary cutoff max(cleaned mean, 0.05 ppm); $n = 135$ buffer-similar systems from `CSP_UBQ_ph0.5_temp5C.csv` with pipeline outputs). Mean = 0.0804 ppm. This figure shows our mean CSP threshold is 0.08 ppm, corresponding to ∼50 Hz at 600 MHz, which is 2–5× larger than the corresponding proton resonance linewidths.



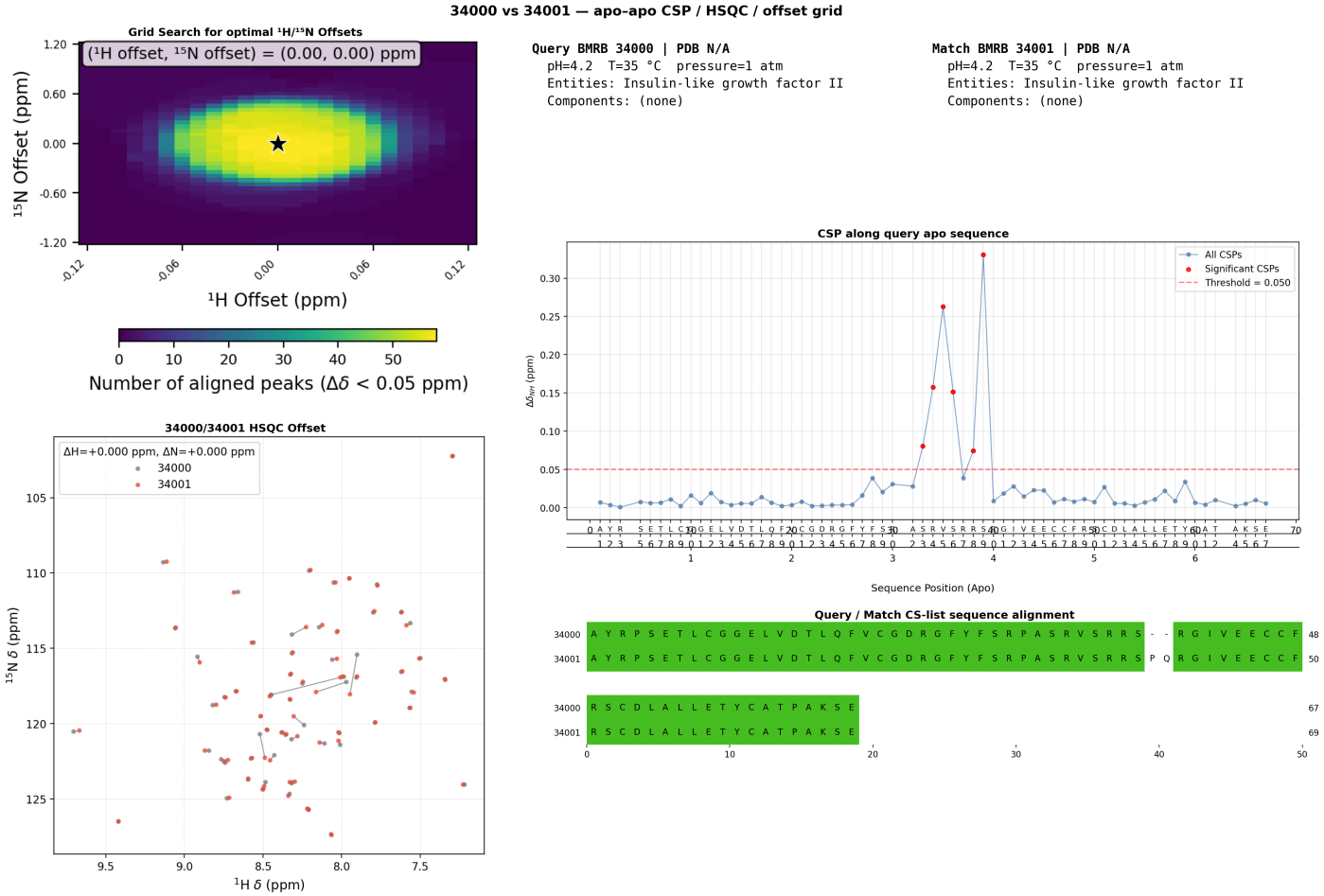
SI Fig. S24. Figure 3-style CA-distance histograms under alternate HN CSP significance thresholds ($n = 135$ buffer-similar systems from `CSP_UBQ_ph0.5_temp5C.csv` with pipeline outputs). Each column is a 3a (TP/FP among significant CSPs) | 3b (full confusion) pair. Thresholds (left to right, then continuing as labeled on the figure): **primary** max(cleaned mean, 0.05 ppm) after iterative outlier removal; **cleaned mean** +1 **SD** and +2 **SD**; **top 10%** and **top 5%** of HN CSP values; fixed cutoffs **0.03**, **0.05**, and **0.10 ppm**; **raw mean**, **raw mean** +1 **SD**, and **raw mean** +2 **SD** (no outlier removal); and **cleaned mean** without the 0.05 ppm floor. This figure reveals our results are relatively stable if we use other reasonable thresholds of what is considered a significant CSP. Some thresholds from this figure we determine are too high (e.g. Raw mean + 1 SD, Raw mean + 2 SD, Top 10%, Top 5%) which results in less distal CSPs than is expected from the other thresholding approaches. SI Fig. S17C shows the continuous ROC analysis, and SI Fig. S17D shows the significant-CSP magnitude distributions.



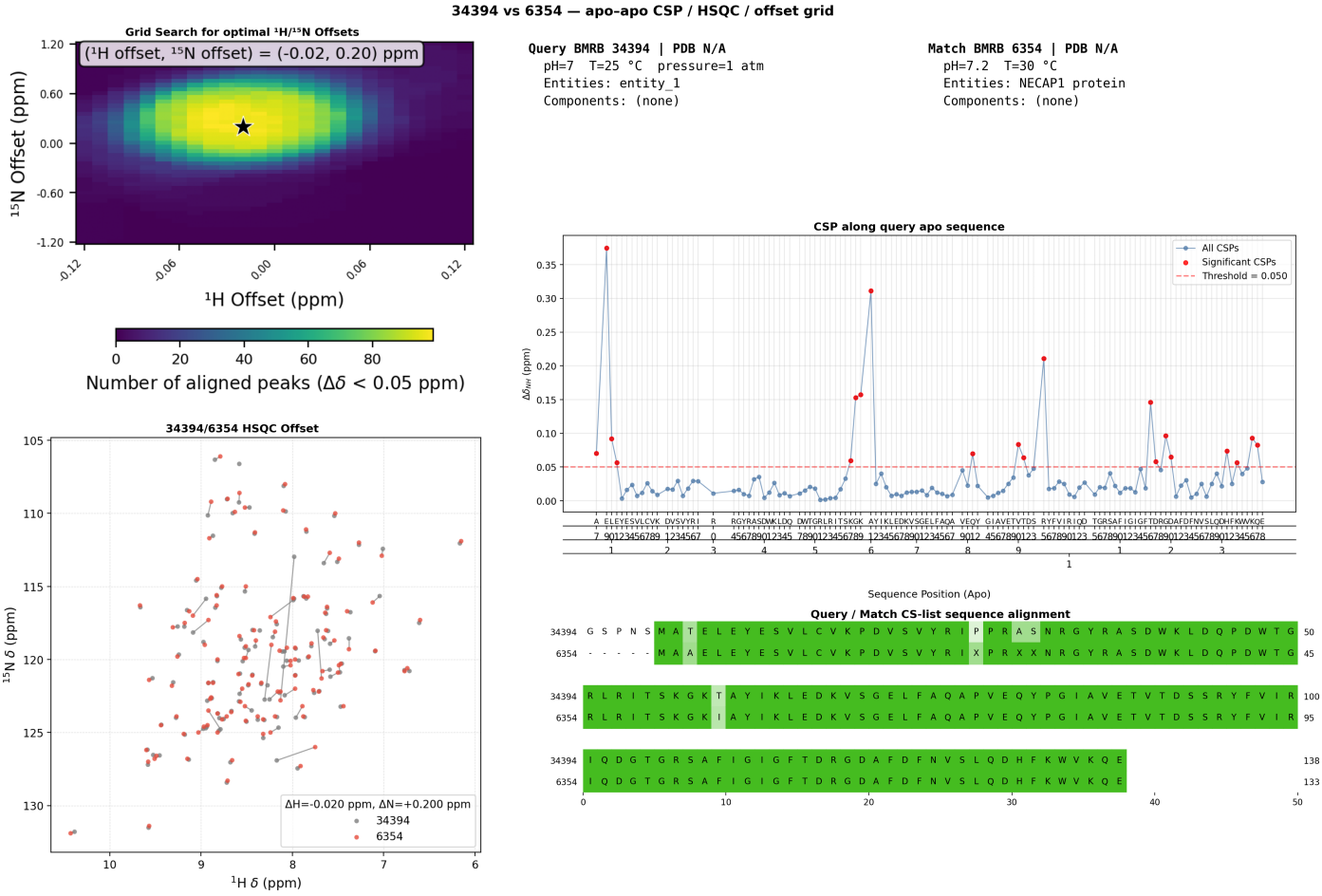
SI Fig. S25A. Apo vs apo control study for BMRB 52080 and 52079. This case involves the CIN85 SH3B–SH3C linker (residues 163–333). There is a sequence mismatch at R227A (BMRB 52080 is the R229A variant; BMRB 52079 is R227A/R229A); BMRB 52079 also has an extra N-terminal Gly. 99.3% of the CSPs for this control are below 0.05 ppm.



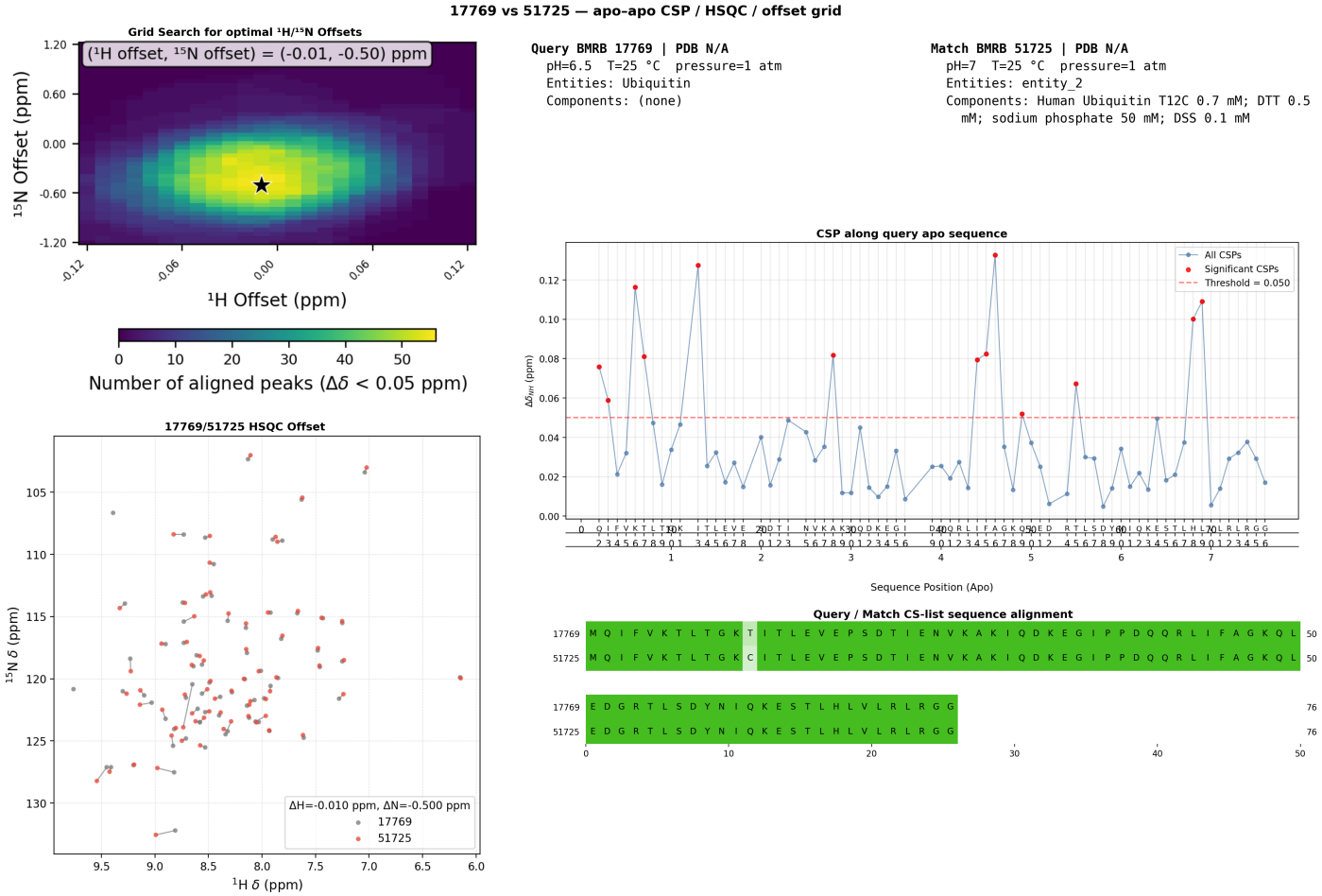
SI Fig. S25B. Apo vs apo control study for BMRB 28070 and 28071. This case involves the SH2 domains of SHP2. BMRB 28071 is the tandem nSH2–cSH2 construct and includes a large N-terminal region (the nSH2 domain and inter-SH2 linker, ∼100 residues) that is absent from the isolated cSH2 entry (BMRB 28070). 95.4% of the CSPs for this control are below 0.05 ppm.



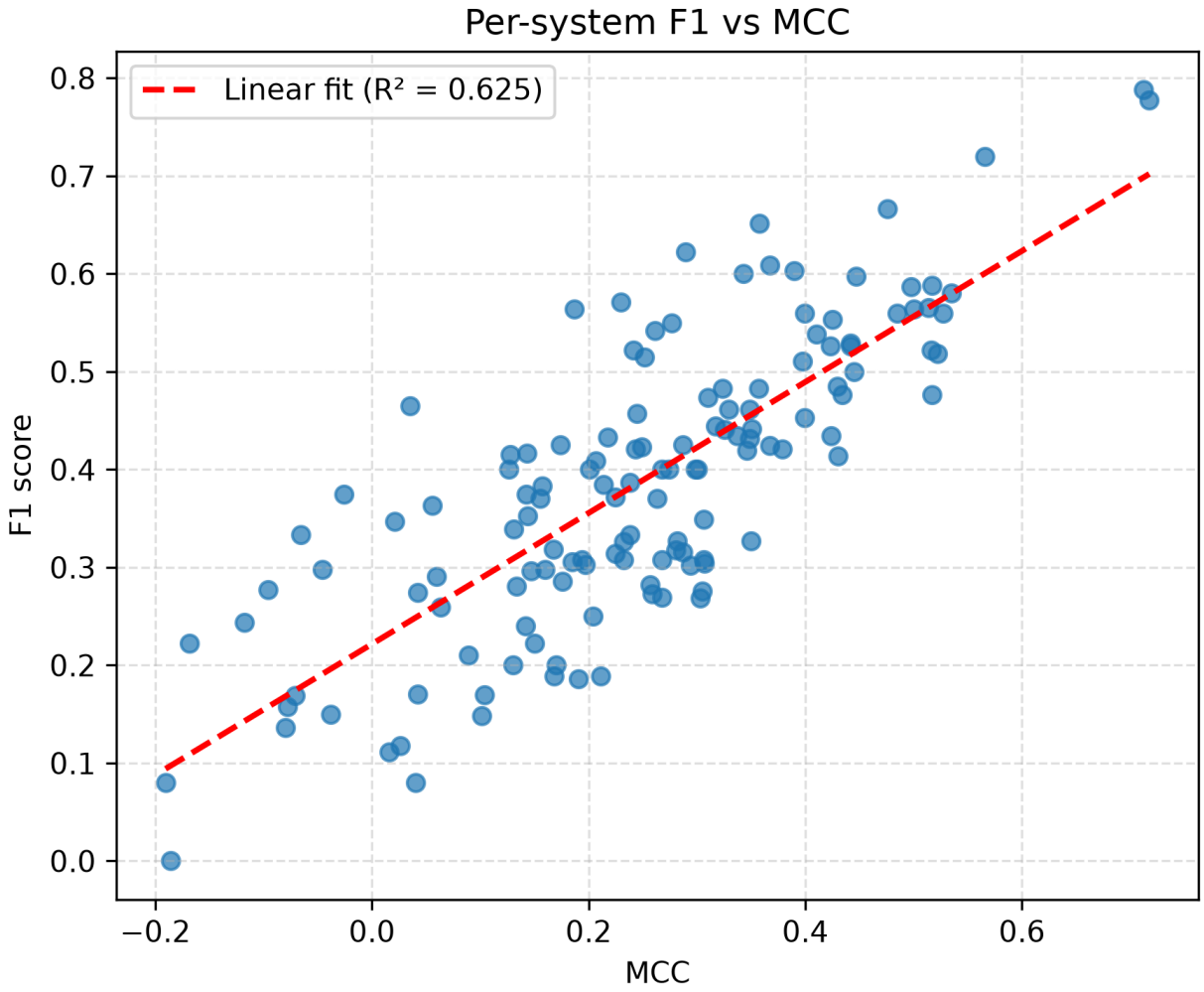
SI Fig. S25C. Apo vs apo control study for BMRB 34000 and 34001. This case involves insulin-like growth factor II (IGF-II). BMRB 34001 is the [S39_PQ]-IGF-II variant and contains a two-residue Pro–Gln insertion after Ser39 relative to wild-type IGF-II (BMRB 34000). 90.6% of the CSPs for this control are below 0.05 ppm.



SI Fig. S25D. Apo vs apo control study for BMRB 34394 and 6354. This case involves the NECAP1 PHear domain. BMRB 34394 has a five-residue N-terminal extension (GSPNS) relative to BMRB 6354, plus substitutions T8A and K59I and short internal insertions (Pro28 and Ala31–Ser32) in the 34394 construct. 83.2% of the CSPs for this control are below 0.05 ppm.



SI Fig. S25E. Apo vs apo control study for BMRB 17769 and 51725. This case involves ubiquitin. There is a sequence mismatch at position 12 (T12C; wild-type Thr in BMRB 17769, Cys in BMRB 51725). 81.2% of the CSPs for this control are below 0.05 ppm.



SI Fig. S26. F1 vs MCC across the $n = 135$ buffer-similar systems in `CSP_UBQ_ph0.5_temp5C.csv` with pipeline outputs. This figure reveals a strong correlation between F1 and MCC measures against the $n = 135$ subset. F1 and MCC are both useful metrics to gauge at a glance the extent of distal CSPs are present in the receptor-ligand system[11].

$$\mathrm{CSP} = \sqrt{\tfrac{1}{2}\left(\Delta\delta_{\mathrm{H}}^2 + (0.14\,\Delta\delta_{\mathrm{N}})^2\right)}$$

SI Equation S1. Combined HN chemical-shift perturbation

$$\mathrm{CSP} = \sqrt{\tfrac{1}{3}\left(\Delta\delta_{\mathrm{H}}^2 + (0.14\,\Delta\delta_{\mathrm{N}})^2 + (0.3\,\Delta\delta_{\mathrm{C}\alpha})^2\right)}$$

SI Equation S2. Combined HN+Cα chemical-shift perturbation

$$\mathrm{CSP} = \sqrt{\tfrac{1}{2}\left(\Delta\delta_{\mathrm{H}\alpha}^2 + (0.3\,\Delta\delta_{\mathrm{C}\alpha})^2\right)}$$

SI Equation S3. Combined Hα+Cα chemical-shift perturbation

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